MILITARY SPECIFICATION

POWER SUPPLY, UNINTERRUPTIBLE, SOLID STATE

(NAVAL SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a 60-hertz (Hz) solid state uninterruptible power supply (UPS) (see 6.4) for Naval shipboard use. The UPS is also referred to as a no-break power supply.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS.

FEDERAL
PPP-C-1842 - Cushioning Material, Plastic, Open Cell (For Packaging Applications).
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY
MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-117 - Bags, Sleeves and Tubing - Interior Packaging.
MIL-S-901 - Shock Test, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915 - Cable and Cord Electrical, For Shipboard Use General Specification For.
MIL-C-915/30 - Cable, Electrical, 1000 Volts, Type TSGU.
MIL-E-917 - Electric Power Equipment, Basic Requirements For (Naval Shipboard Use).
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-L-3661 - Lampholders, Indicator Lights, Indicator-Light Housings, and Indicator-Light Lenses, General Specification for.
MIL-M-7793 - Meter, Time Totalizing.
MIL-Q-9858 - Quality Program Requirements.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
DOD-B-15072 - Batteries, Storage, Lead-Acid, Portable; General Specification for (Metric).
MIL-F-15160 - Fuses, Instrument, Power, and Telephone - General Specification for.
MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).
MIL-M-16034/9 - Voltmeters, A.C. and D.C., Portable, 0.25-Percent Accuracy Class.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, SEC 6124, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY (Continued)

MIL-F-16552 - Filters, Air Environmental Control System, Cleanable, Impingement (High Velocity Type).
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated Housing (Shipboard Use).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-C-17587 - Circuit Breakers, Air, Electric, Open Frame Removable Assembly (Shipboard Use).
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-M-24365 - Maintenance Engineering Analysis: Establishment of, and Procedures and Formats for Associated Documentation; General Specification for.
MIL-M-28787 - Modules, Electronic, Standard Hardware Program, General Specification for.
MIL-C-45662 - Calibration System Requirements.
MIL-H-46855 - Human Engineering Requirements for Military Systems, Equipment and Facilities.
MIL-B-81705 - Barrier Materials, Flexible, Electrostatic-Free, Heat Sealable.
MIL-P-81997 - Pouches, Cushioned, Flexible, Electrostatic-Free, Reclosable, Transparent.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-415 - Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-471 - Maintainability Verification/Demonstration/Evaluation.
MIL-STD-721 - Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production.
MIL-STD-882 - System Safety Program Requirements.
MIL-STD-965 - Parts Control Program.
MIL-STD-1310 - Shipboard Bonding, Grounding and Other Techniques for Electromagnetic Compatibility and Safety.
MIL-STD-1326 - Test Points, Test Points Selection and Interface Requirements for Equipments Monitored by Shipboard On-Line Automatic Test Equipment.
MIL-STD-1375 - Provisioning, Initial Support General Requirements For.
MIL-STD-1378 - Requirements for Employing Standard Hardware Program Modules.
DOD-STD-1399, Section 300 - Interface Standard for Shipboard Systems, Electrical Power, Alternating Current (Metric).
MS90363 - Box, Fiberboard with Cushioning Insert, Limited Re-Use (For Items 10 Pounds or Less).

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

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DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS S3801-1170895 - Air Filters, Navy Standard.
NAVSHIPS 9000-S6202-73980 - Electric Plant Installation Standard Methods.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSEA 0967-LP-312-8010 - Maintainability Design Criteria Handbook for Design of Shipboard Electronic Equipment.
NAVSEA 0967-LP-597-1010 - Electronic Equipment Parts Application and Reliability Information.

NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)
Coordinated Shipboard Allowance List, Portable Electrical Testing and Measuring Equipment, Part II B Group II Section B - Portable Electrical Testing and Measuring Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
Y32.16 - Reference Designations for Electrical and Electronics Parts and Equipments.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample of a solid state UPS shall be examined and tested as specified in 4.4 (see 6.3).

3.2 Basic configuration. The major subassemblies of the UPS shall include an input contactor, a rectifier-charger, a storage battery, a static inverter, an automatic transfer static-switch, and a bypass switch. The basic UPS configuration shall be as shown in figure 1.

3.3 Performance requirements.

3.3.1 Input power. The rated input for a UPS having a power rating 3 kilowatts (kW) or less shall be 115 volts, 60-Hz, single-phase, ungrounded, type I power complying with the steady-state and transient characteristics and ranges as specified in DOD-STD-1399, Section 300. The rated input for a UPS having a power rating above 3 kW shall be 440-volt, 60-Hz, 3-phase, ungrounded, type I power complying with the steady-state and transient conditions and ranges as specified in DOD-STD-1399, Section 300.

3.3.1.1 Self-protection from input voltage variation and transient. When the UPS is supplying any load from no-load to rated-load, the UPS shall withstand (without damage or blowing fuses) the following conditions of input power:

(a) A loss of input power for any time duration.
(b) Any input voltage or frequency or both outside the transient limits specified in DOD-STD-1399, Section 300, as follows:
    (1) The input voltage instantly drops to zero.
    (2) The input voltage drops to between 0 and 82 percent of rated voltage and input frequency drops to between 0 and 92 percent of rated frequency. The voltage-frequency relation during the period will be such that voltage will be maintained at rated voltage until the frequency drops down to about 83 percent of rated

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IT FOR TABLE FIGURE 5. Reverse voltage protection test circu

frequency, then the voltage drops in proportion to the frequency, and finally the input voltage collapses to zero when the frequency is about 35 percent of rated frequency.

(3) The input voltage rises to 150 percent of rated voltage, or input frequency rises to 125 percent of rated frequency, or both. Upon return of the input voltage and the frequency to the specified range (see 3.3.1), the rectifier-charger shall automatically restart (without manual restart) in its normal starting sequence to operate and supply the batteries and inverter. Upon return of the input power, the UPS output voltage and frequency shall be maintained within the regulation band as specified in 3.3.2.1 and 3.3.2.6, respectively. The performance of the equipment shall meet all the requirements of this specification without any adjustment or replacement of parts.

(4) The input voltage phase rotation is in a reverse order (AC-CB-BA).

3.3.1.2 Self-protection from input voltage spikes. When the UPS input voltage experiences a transient (line-to-line, or line-to-ground, or both) up to 2500 volts having a basic impulse level (BIL) wave shape as shown on figure 2, the UPS output voltage waveform shall conform to 3.3.2.8. The UPS shall continue to operate without shutting down or malfunctioning. The calculated 60-Hz source surge impedance (the total source surge impedance is considered to be the cable impedance only) will be as specified below:

Navy type cable	Current rating of cable in amperes at 50°C ambient	Surge impedance in ohms
TSGU 14	47	26
TSGU 23	64	26
TSGU 50	101	20
TSGU 75	136	20
TSGU 100	160	19
TSGU 150	216	18
TSGU 200	250	18
TSGU 300	320	18
TSGU 400	400	17

3.3.1.3 Input current harmonics. When the UPS is powered by a 60-Hz sinusoidal voltage within the range specified in 3.3.1 and with the UPS supplying any load from no-load to rated-load, the operation of the UPS shall not result in input current harmonics exceeding those shown on figure 3.

3.3.1.4 Input current balance. When a 3-phase UPS (see 3.3.1) is supplying any load from 10 percent to rated-load, and the input voltage unbalance is 3 percent as specified in DOD-STD-1399, Section 300, the input current shall be balanced so that the current in any line of the input shall not differ by more than 5 percent of the rated input current from the arithmetic average current in the three lines.

3.3.1.5 Starting time. The UPS shall be capable of accepting any load from no-load to rated-load and operating in compliance with all requirements specified herein in not more than 30 seconds after close of the input contactor. The UPS shall not require adjustments to meet these requirements when the input voltage and frequency vary within the ranges specified in 3.3.1. Means shall be provided within the UPS to automatically preclude the UPS from accepting any connected load during the starting period.

3.3.1.5.1 Prohibition of start. The UPS shall not be capable of being manually started when the input voltage is less than 103 volts or higher than 127 volts for the single-phase UPS, and less than 396 volts or higher than 484 volts for the 3-phase UPS.

3.3.1.6 Starting input current. Input current to the UPS shall be limited during the starting time (see 3.3.1.5). The peak input starting (or inrush) current shall be not greater than 150 percent of the peak full-load steady-state input current. An auxiliary contactor may be used as a start contactor provided that the start contactor is not required for normal operation.

3.3.1.7 No-load input current. When the UPS storage battery is fully charged, the maximum no-load input current shall be not greater than 16 percent of rated-load input current.

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3.3.1.8 Grounding. The UPS shall operate to meet all requirements specified herein when connected to an ungrounded ship's input power system having 3 to 20 microfarad capacitance from any line to ground or with any single input or output line intermittently or permanently grounded.

3.3.2 Output voltage. Rated output voltage of the UPS of a power rating 3 kW or less shall be 115-volt, 60-Hz, single-phase, ungrounded. Rated output voltage of the UPS of a power rating above 3 kW shall be 440-volt, 60-Hz, 3-phase, ungrounded. Both the input and the output phase rotation of a 3-phase UPS shall be a clockwise sequence of AB-BC-CA.

3.3.2.1 Voltage regulation. The average of the root-mean-square (rms) values of the output voltage shall be maintained at the regulated point (at the remote sensing point or at the UPS output terminals (see 3.3.2.3)) within plus or minus 5 percent of the rated output voltage under the following variable conditions and any combination thereof at all output voltage settings within the adjustment range specified in 3.3.2.4:

(a) A voltage drop from the terminals of the UPS to the remote sensing point of any amount from 0 to 7 percent of rated output voltage.
(b) A load variation from no-load to 200 percent of rated-load.
(c) Load power factor (p.f.) varying between 0.6 lagging and 0.9 leading.
(d) An input voltage and frequency variation within the range specified in 3.3.1 with the storage battery connected and disconnected.
(e) Any input voltage and frequency variation, even outside the range specified in 3.3.1, but with the storage battery connected and charged.
(f) Variation of the UPS temperature from cold start (at ambient temperature ranging from 0°C to 50°C) to maximum operating temperature (at ambient temperature ranging from 0°C to 50°C).
(g) Under the unbalanced load condition specified in 3.3.2.11.
(h) Under the non-linear load conditions specified in 3.3.2.2.
(i) With batteries supplying the UPS output power for a period up to 15 minutes.
(j) When batteries are recharged as specified in 3.3.12.7.

3.3.2.2 Non-linear loads. Typical non-linear loads are defined as loads developed by single-phase or 3-phase full-wave rectifying systems, and as such will have maximum harmonic current content imposed on the UPS output line in accordance with the following formula when the load system is supplying resistive loads:

$$I_N = \frac{100}{N} \text{ percent of fundamental current}$$

Where:
I_N is the Nth harmonic current and N is the harmonic number. N cannot be any multiple of 3. When a non-linear load up to 100 percent of the rated-load is applied to the UPS, the output voltage waveform shall remain within the limits as specified in 3.3.2.8, the output voltage regulation shall be in accordance with 3.3.2.1, and the amplitude modulation shall be as specified in 3.3.2.7.

3.3.2.3 Remote sensing. The UPS shall operate with the output voltage sensing at a remote sensing point. The remote voltage sensing point will be connected to the unit by a type TSGU-3 in accordance with MIL-C-915/30 or equivalent cable. The cable will be furnished by the installation activity. The maximum length of the remote sensing cable shall be 200 feet. For any loads between no-load and 200 percent of rated-load (see 3.3.6), the voltage at the remote sensing point shall be automatically maintained at the adjusted voltage (see 3.3.2.4) and shall conform to all steady-state voltage regulation, voltage modulation, waveform, and nonlinear loads requirements (see 3.3.2.1, 3.3.2.7, 3.3.2.8 and 3.3.2.2, respectively) specified herein. The voltage drop between the terminals of the UPS and the remote voltage sensing point shall not exceed 7 percent of rated voltage at 200 percent of rated-load. In the event that the remote voltage sensing signal is disconnected or otherwise interrupted, the UPS shall automatically shift the voltage sensing from the remote point to the UPS output terminals. Under these conditions, the voltage at the UPS terminals shall be the same as or not greater than 2 percent above the adjusted voltage level which existed at the remote point prior to transfer. After restoration of the remote voltage sensing signal, the voltage regulation shall automatically shift back to the remote sensing point.

3.3.2.4 Voltage adjustment. Means shall be provided to adjust the output voltage at the regulated point over a range of plus or minus 5 percent of the rated voltage, with adjustment resolution not more than 100 millivolts at any load from no-load to 200 percent

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rated-load under any steady-state input conditions. The adjustment device shall have a slotted shaft for screwdriver type adjustment and shall be equipped with a locking device, such as a lockout. The end of the slotted shaft shall be covered by an acorn nut or spring-return cover to prevent tampering.

3.3.2.5 Transient output voltage and recovery. A sudden change of any load on the UPS from no-load to full-load, 0.8 p.f. lagging, and from full-load 0.8 p.f. lagging to no-load shall cause an output voltage deviation of not more than plus or minus 18 percent of rated voltage. The output voltage shall recover and stay within the steady-state regulation band within 0.05 second. Voltage transients developed within the UPS shall not reflect back into the input power system such that transient limits of DOD-STD-1399, Section 300 are exceeded for alternating current (a.c.) input (type I power).

3.3.2.6 Frequency regulation. Output frequency shall be maintained within plus or minus 5 percent of rated frequency under the following conditions and combination thereof:

(a) A load variation from no-load to 200 percent of rated-load.
(b) Load p.f. varying between 0.6 lagging and 0.9 leading.
(c) An input voltage and frequency variation within the range specified herein.
(d) Variation of the UPS temperature from cold start (at ambient temperature ranging from 0°C to 50°C) to maximum operating temperature (at ambient temperature ranging from 0°C to 50°C).
(e) Variation of output voltage specified in 3.3.2.4.
(f) Any load changes as specified in 3.3.2.5.

3.3.2.7 Amplitude and frequency modulation. Amplitude and frequency modulation of the output voltage (as defined in DOD-STD-1399, Section 300) shall not exceed 2 percent and 0.5 percent, respectively, under any of the conditions or combinations of conditions of 3.3.2.1.

3.3.2.8 Output voltage waveform. Under any load conditions between no-load and rated-load, 0.8 lagging p.f. and unity p.f. (see figure 4), the rms value of the sum of all of the harmonic content in the output voltage shall not exceed 5 percent of the rated output voltage. The rms value of any single harmonic shall not exceed 3 percent of the rms value of the rated output voltage. The deviation factor (as defined in DOD-STD-1399, Section 300) of the output voltage waveform shall not exceed 5 percent. Amplitude of voltage spikes of any frequencies of 20 kilohertz (kHz) and above shall not exceed 4 percent of the rms value of the rated output voltage.

3.3.2.9 Output phase displacement. For 3-phase UPS, the output voltage displacement angle between adjacent voltage phases shall not exceed 120 ± 1 degrees under all conditions of operation specified herein.

3.3.2.10 Output voltage balance under balanced load conditions. For the 3-phase UPS, the difference between phase voltages at the voltage regulated point shall be not greater than 3.0 percent of rated voltage at no-load and at any balanced 3-phase load up to rated current with any p.f. between 0.6 lagging and 0.9 leading.

3.3.2.11 Output voltage balance under unbalanced load conditions. For the 3-phase UPS, the voltage variation between phases at the voltage regulated point shall be not greater than 3 percent of rated voltage under the unbalanced load conditions shown in table I. The UPS shall operate with a single phase 0.6 p.f. load carrying a current 15 percent of rated current. The voltage of the loaded phase at the voltage regulated point shall have regulation as specified in 3.3.2.1.

TABLE I. Unbalanced loads.

Load conditions	Line A percent of rated current	Line B percent of rated current	Line C percent of rated current	P.f.
1	90	75	75	0.6 lagging to unity
2	75	90	75	0.6 lagging to unity
3	75	75	90	0.6 lagging to unity

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3.3.2.12 Input voltage change effect. When the UPS is supplying any load from no-load to rated-load at 0.8 lagging and unity p.f. and with any output and input steady-state voltage and frequency within the range specified herein, the occurrence of any input transient voltage up to 18 percent with recovery time of 2 seconds as specified for type I power in DOD-STD-1399, Section 300, shall not cause an output voltage drop of greater than 5 percent of the initial value. It shall recover to and remain within one percent of the initial value within 2 seconds from the initiation of the input voltage change.

3.3.3 Ratings. The ratings of the UPS shall be 1, 2, 3, 5, 10, 15, and 30 kW at 0.8 lagging p.f., as specified (see 6.2.1). The kilovolt-ampere (kVA) rating and p.f. rating for each kW rating shall be within the power range (shaded area) as shown on figure 4. The UPS shall be designed to meet all requirements specified herein when operating within this range.

3.3.3.1 Duty. The UPS shall operate continuously at the specified power rating within the p.f. range as shown on figure 4.

3.3.3.2 Motor starting. The UPS shall be capable of starting a 60-Hz motor having a starting current up to 200 percent of the rated UPS output current. The motor starting p.f. will be 0.3 lagging, and the starting current will last for 30 seconds. During the motor starting for that portion of load cycle when the p.f. of the total load is between 0.8 lagging and unity, the output transient voltage of the UPS shall be as specified in 3.3.2.5.

3.3.4 Efficiency. When the battery is disconnected, the overall efficiency of the UPS at the rated input voltage and frequency (see 3.3.1) shall be not less than 80 percent at rated output at 0.8 p.f. and unity p.f. The efficiency at 50 percent rated output at 0.8 p.f. and unity p.f. shall be not less than 70 percent. Losses at other percent load shall be not greater than that shown on figure 5.

3.3.5 No-load losses. When the battery is disconnected, the no-load losses of the UPS shall be not more than 80 percent of the rated-load losses at rated input voltage and frequency.

3.3.6 Overload. The UPS shall be capable of providing and sustaining an overload current of 200 percent of rated current at the rated output voltage for 1 minute and meet all the requirements specified herein (see figure 6). When the 200 percent current lasts more than 1 minute, the load shall be automatically transferred from the UPS to the input a.c. lines. An indicating light shall be provided to show overload transfer. Terminals shall be provided for a remote overload transfer indicator. The output voltage regulation for any overload current up to 200 percent of the rated output current shall be as specified in 3.3.2.1.

3.3.7 Fault and malfunction protection.

3.3.7.1 Fault isolation. The UPS shall incorporate fault isolation and malfunction protection for each module or drawer to prevent cascading to other modules or drawers. A shorted or non-commutated power semiconductor shall not result in damage to any other part of the UPS, with the exception that a single fuse may blow.

3.3.7.2 Fault current. Output current between 200 percent and 230 percent of rated current is considered fault current. The UPS shall be capable of providing and sustaining any fault current up to 230 percent of rated current for 10 seconds when a fault develops at any external circuits supplied by the UPS. After the UPS has supplied the fault current for 10 seconds, the UPS shall automatically transfer its load(s) to the input a.c. line. When the fault is cleared, the load(s) shall automatically be transferred to the UPS without disturbance to the load(s).

3.3.7.3 Short circuit protection. Output current above 230 percent of rated current is considered short circuit current. The UPS shall protect itself without damage to any parts from any short circuit on its output. Protection shall be provided by current limiting to a value not less than 230 percent or greater than 250 percent of rated current (see figure 6). The current limiting shall not be applied to any output current less than 230 percent of rated current. On failure of the current limiting circuit, the UPS shall be protected by fuses (see 3.3.7.4). After the UPS has supplied short circuit current for 10 seconds, the UPS shall automatically transfer its load(s) to the input a.c. lines. When the short circuit is cleared, the load(s) shall automatically be transferred to the UPS without disturbance to the load(s).

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3.3.7.4 Internal circuit protection. Fuses shall be provided for all power circuits and semiconductors as back up for the current limiting protection (see 3.3.7.3). Each drawer containing power switching semiconductors shall be separately fused. Circuits with filter capacitors shall have fuse protection against capacitor failure.

3.3.8 Parallel operation. The UPS shall provide paralleling capability at the levels of reliability and maintainability specified herein. It shall be possible to parallel up to the maximum of three UPS without interrupting the load and without introducing transient voltages at the input line and output load bus outside the specified limits. In parallel operation, the internal oscillator of one of the UPS shall be automatically selected as the master oscillator and synchronized to the input a.c. lines. The internal oscillators of the other two UPS shall be automatically slaved to the "master oscillator". The frequency of the "master oscillator" shall be continuously monitored to enable one of the other oscillators to be automatically substituted whenever the "master oscillator" drifts out of tolerance or malfunctions. The paralleling and synchronization shall be automatically accomplished.

3.3.8.1 Real and reactive load division. When paralleling identical rated kW UPS units, the maximum difference in kW supplied by the units in parallel shall not exceed 10 percent of the kW rating of one of the UPS under all load conditions between 10 and 100 percent of the total rated load of the paralleled UPS. The maximum difference in kilovar (kvar) supplied by the UPS operating in parallel shall not exceed 10 percent of the kvar rating of one of the UPS under any load condition between 10 and 100 percent of the total rated-load of the paralleled UPS. When paralleling different rated kW UPS, the output of each UPS shall be proportional to the individual size. The maximum difference in kW supplied by the UPS in parallel shall not exceed 10 percent of the average kW rating of all UPS in parallel under all load conditions between 10 and 100 percent of the total rated-load of the parallel UPS. The maximum difference in kvar supplied by the UPS operating in parallel shall not exceed 10 percent of the average kvar rating of all UPS in parallel under any load condition between 10 and 100 percent of the total rated-load of the parallel UPS. The above requirements shall apply for all input voltage variations specified in 3.3.1 and other environmental requirements specified herein.

3.3.8.2 Selective tripping. When any malfunction occurs to any one of the UPS operated in parallel, the malfunctioned UPS shall automatically shut itself down without causing the remaining operating UPS to shut down.

3.3.9 Automatic transfer switch. An automatic transfer static switch shall be provided and connected between the static inverter and the load(s) as shown in figure 1. The normal supply to the transfer switch shall be from the static inverter and the alternate supply shall be the input a.c. lines to the UPS. When the voltage and frequency of the input a.c. lines are within the steady-state range specified in 3.3.1, the transfer switch shall sense the status of the UPS and make the automatic transfer to the load(s) from the normal supply to the alternate supply when any of the following conditions exist:

(a) The UPS output voltage at the regulated point is outside the plus or minus 5 percent limits as specified in 3.3.2.1.
(b) The UPS output frequency is outside the plus or minus 5 percent limits as specified in 3.3.2.6.
(c) The loss of direct current (d.c.) input to the inverter.
(d) Any output overload, fault current, or short circuit conditions as specified in 3.3.6, 3.3.7.2, and 3.3.7.3.

The transfer of load(s) from the normal supply to the alternate shall also automatically disconnect the static inverter from the load(s). The load transfer shall be accomplished in no more than 8 milliseconds (ms). One second after the condition which initiated the transfer has been corrected and the normal inverter output has been verified, the transfer circuit shall automatically synchronize the UPS to the input a.c. lines and retransfer the load(s) back to the normal supply without disturbance to the load(s). When the UPS is supplying power to the load(s) through the transfer switch, the UPS output shall meet all requirements specified herein.

3.3.10 By-pass switch. A by-pass switch shall be provided for the transfer of the load(s) from the inverter to the input a.c. lines, or vice versa. The load transfer shall be accomplished in less than one second and shall have no disturbance to the load. The output voltage shall meet all requirements specified herein during the transfer operation.

3.3.11 Synchronization. When the output load(s) of the UPS is supplied by the normal supply (see 3.3.9), the UPS output voltage phase angle shall be synchronized to within plus or minus 5 electrical degrees of the input a.c. line voltage phase angle. The synchronization shall be activated by a contact of the by-pass switch as shown in figure 1. When

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the a.c. input deviates beyond the type I power frequency regulation band of plus or minus 5 percent, or the by-pass switch contact is opened, the UPS shall revert to its internal reference frequency and shall continue to maintain the output frequency within the steady-state regulation limits specified in 3.3.2.6. After the UPS input power returns to the steady-state regulation range, and the by-pass switch contact reclosed, the UPS output frequency shall be resynchronized to the input a.c. voltage. The rate of the frequency change during the resynchronizing period shall be 1 Hz per second.

3.3.12 Rectifier-charger. The rectifier-charger shall provide power for the UPS output and to charge the battery bank.

3.3.12.1 Rectifier-charger operation. When voltage and frequency of the input power to the UPS is interrupted or experiences an out of the type I power steady-state tolerance condition, the rectifier-charger may shut down and the batteries shall automatically supply power to the inverter. When the input power to the UPS is restored and within the type I power steady-state tolerance, the rectifier-charger shall automatically re-energize, recharge the batteries, and provide a constant float charge to these batteries between the input power interruptions without disturbance in the inverter output. The rectifier-charger shall be capable of recharging the batteries to a fully charged condition while simultaneously providing power to the inverter for all operating conditions of the UPS specified herein.

3.3.12.2 Battery charging circuit. The battery charging circuit shall automatically regulate the rectifier output voltage and current to such value as necessary to fully recharge the completely discharged battery bank. The shall not be any reverse current flow from the battery bank when the UPS is disconnected from the input power.

3.3.12.3 Battery electrolyte temperature. When charging the batteries, the temperature of the electrolyte shall not exceed 125°F. When this temperature is reached, the rate of charge shall be reduced or the charge discontinued until the battery electrolyte has cooled off.

3.3.12.4 Low battery voltage shutdown. If the UPS input voltage fails for a period such that the battery discharges to the point where the UPS output voltage reduces to 95 percent of rated, the UPS shall automatically shutdown to prevent overdischarge of the batteries.

3.3.12.5 Battery float charge. The constant voltage float charge supplied by the rectifier-charger will not excessively overcharge the batteries so as to cause electrolyte spillage, gassing and high temperature of the batteries.

3.3.12.6 Rectifier-charger protection. The rectifier-charger shall be protected by limiting its output current and voltage.

3.3.12.7 Rectifier-charger and battery duty cycle. The rectifier-charger and the battery shall be capable of operating with the following duty cycle:

(a) When power to the UPS is interrupted, the batteries shall supply power to the inverter for 15 minutes of rated-load operation.
(b) The rectifier-charger shall re-energize automatically and simultaneously charge the batteries and supply power to the inverter within one minute after the input power to the UPS is restored and within the type I power steady-state limits.
(c) The rectifier-charger shall simultaneously charge the batteries and supply the inverter at any load from no-load to rated-load as shown in figure 4 for one hour.
(d) The rectifier-charger and the battery shall be capable of repeating the cycle steps (a) and (b) immediately following cycle step (c).
(e) After (d), the rectifier-charger shall be capable of fully charging the batteries within eight hours while supplying the inverter operating at rated-load. During the above duty cycle, the UPS output shall be maintained and shall meet all requirements specified herein.

3.3.13 Battery bank. The batteries shall be lead-acid storage type in accordance with DOD-B-15072.

3.3.13.1 Battery duty cycle. The battery duty cycle shall be as specified in 3.3.12.7.

3.3.13.2 Battery output voltage. The UPS shall meet all requirements specified herein when the d.c. link (see figure 1) voltage is the same as or up to 2 volts less than the battery terminal voltage. The maximum voltage drop in the cable connecting the battery and the d.c. link shall be 2 volts under all input and output conditions.

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3.3.13.3 Battery configuration. Batteries shall be supplied on enclosed racks. The battery rack and associated enclosure shall conform to Drawing 9000-S6202-73980, Section 3, Sheets 111, 112, 113, and 114. The battery rack and enclosure shall be separated from the other cabinets (see 3.4.3.2) and shall be properly ventilated to prevent accumulation of hydrogen gas. An individual independent battery bank shall be supplied for each UPS such that each d.c. bus is isolated from all others.

3.3.14 Inverter. The inverter shall maintain the UPS output voltage and frequency within the steady-state limits as specified in 3.3.2.1 and 3.3.2.6 when its input is transferred between the batteries and the rectifier-charger.

3.3.15 Leakage current. The rms leakage current caused by the EMI filter capacitor connected from each line to the ship's ground shall not be greater than 30 milliamperes per line for each input and output line of the UPS.

3.4 Design and construction.

3.4.1 Basic requirements. In addition to the requirements of this specification, the UPS basic design shall be in accordance with the requirements of MIL-E-917, except the following paragraphs and all subparagraphs pertaining thereto. When a requirement of MIL-E-917 conflicts with a requirement of this specification, the requirement of this specification shall govern.

(a) Fundamental equipment characteristics (excluding safety).
(b) Grounding.
(c) Input power variations (a.c. equipment).
(d) Inclined operation.
(e) Ambient temperature.
(f) Impact shock and vibration.
(g) Equipment mounting.
(h) Radio interference.
(i) Noise reduction for submarine.
(j) Piping system.
(k) Drawings.
(l) Manuals.
(m) Workmanship.

3.4.2 Electrical designs. The UPS shall utilize building block module concept with solid state circuitry and devices only. Electromechanical type relays may be used only under the conditions applicable to the use of non-standard parts (see 3.5.1). Vacuum tubes or other similar thermionic devices shall not be used. Control and logic circuits shall have a common negative. Control wires shall be separated from the power conductors as required to avoid noise pickup in the control circuits. Optical coupling may be used to effect electrical isolation between control circuits and power semiconductor devices.

3.4.2.1 Dielectric withstanding voltage and insulation resistance. The UPS shall pass insulation resistance and dielectric withstanding voltage tests specified in 4.7.3 and 4.7.4. Insulation resistance of the circuits shall be not less than 10 megohms when measured with the UPS at ambient temperature of 25°C + 2°C. This requirement shall be met without resorting to either disassembly or disconnection of any cabinet, drawer, module, or part. There shall be no interconnection of circuits, the shorting together of terminals, or the short-circuiting, by-passing, or grounding of parts or circuit elements during the test to prevent the test voltage from being applied to parts. Such measures, however, may be employed when specifically authorized in the accepted test procedure for convenience and effectiveness in testing the UPS provided such measures employed do not prevent the test from being effective on portions of the UPS. Use of any such measures in the test procedures shall be entirely at the option of the Naval Ship Engineering Center (NAVSEC) based upon recommendation and justification to be provided by the contractor. The UPS shall enable the required insulation resistance and dielectric withstanding voltage tests on the insulation system to be performed without causing either incipient or catastrophic damage or significant (within acceptable product variability tolerance) degradation of any material or part, including the electrical insulation. Furthermore, a defect in the insulation system shall not cause performance of either of the tests, when properly conducted, to result in damage or degradation to any other material or part except that which possibly may be inflicted to the immediately adjacent parts separated by the defective insulation.

3.4.2.2 Power transformers. When input or output transformers are used, they shall be a balanced arrangement (i.e., three phase Δ connection, three phase Y connection, six phase star connection, etc.).

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3.4.3 Mechanical design. The UPS shall provide front access and maintenance with plug-in, pull-out drawers, with the exception of the batteries. When the weight of a power transformer (either input or output) is heavier than that allowed in one drawer, as specified in NAVSEA 0967-LP-312-8010 (see 3.7.3.2), the power transformer shall be mounted on a roll-out rack for access, removal and replacement. When power transformers are mounted on roll-out racks, the electrical connections to the power transformers shall be made with either plug-in connectors or lug type terminals. Pull-out drawers and roll-out racks shall have stops to lock in the pull-out position. Drawers and roll-out racks shall be interlocked with the power input contactor and battery circuit breaker to prevent opening of the drawers and pull-out racks before interruption of the power to the UPS. Drawers shall be so designed that non-identical drawers cannot be interchangeably plugged in the UPS. The drawer and roll-out rack interlock shall have means to prohibit operation of the UPS when one or more drawers or racks have been removed from the UPS. EMI filters shall be internally mounted in the cabinet. The input contactor may be internally mounted in the cabinet.

3.4.3.1 Size and weight. The UPS shall not exceed the total weight and total volume requirements as shown in tables II and III.

TABLE II. Volume, weight, and number of cabinets.

kW rating	Cabinet		
	Number of cabinets	Maximum total weight (lbs)	Maximum total volume (ft^3)
1	1	1200	24
2	1	1500	24
3	1	1800	24
5	1	2200	24
10	2	2500	48
15	2	3000	48
30	2	4000	48

TABLE III. Weight and dimensions of the battery rack and associated enclosure.

kW rating	Battery rack and associated enclosure			
	Maximum weight (lbs)	Maximum length (inches)	Maximum depth (inches)	Maximum height (inches)
1	1600	80	24	60
2	2800	90	34	60
3	2800	90	34	60
5	4500	140	28	60
10	4700	140	28	60
15	5800	140	36	60
30	15200	160	32	60

3.4.3.2 Cabinet arrangement. The UPS, excluding the battery rack enclosure, shall have not more than the number of cabinets as specified in table II. Each cabinet shall have a height not to exceed 72 inches with appropriate width and depth to enable it to go through a 30 inch by 30 inch hatch with round corners of 7.5 inch radius, and 26 inch by 45 inch doors with round corners of 7.5 inch radius. Drawers may be removed to enable passage through the hatches and doors. Cabinets shall be designed for installation side by side without any separation for ventilation and without having any effect on operating temperatures.

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3.4.3.3 Cabinet enclosure. The UPS shall have a dripproof protected (45-degree) enclosure conforming to MIL-STD-108 and class I of MIL-E-2036. Cable entrance plates to accommodate stuffing tubes (see 3.4.4) shall also be dripproof and shall be on top of the cabinets.

3.4.3.4 Ground connection. Exposed metal parts of the cabinets shall be at ground (ship's hull) potential at all times. Exposed metal portions of electrical parts (such as switches and rheostats) which are not part of the electrical circuits, shall be electrically grounded to the frame of the cabinet. A ground stud shall be provided on each cabinet in accordance with MIL-STD-1310 to accept a class C, type 2 bond strap which will be supplied and installed by the installation activity. The ground stud shall be clearly marked and shall be readily accessible and so located or protected as to reduce the probability of breakage during transit or operation.

3.4.4 Interconnection cables.

3.4.4.1 Interconnection between cabinets (see 3.4.3.2) shall be made using electrical cable or bus bars. Removable plates on the sides and on the top of each cabinet shall be provided for the entrance of the interconnection cables or bus bars.

3.4.4.2 Input and the output cables shall be selected in accordance with MIL-C-915 and shall be specified on the installation drawings (see 3.9.1). These cables shall not be furnished with the UPS. Threaded terminal studs with lockwashers and locknuts shall be provided to accept the input and output cable terminal lugs.

3.4.5 Monitoring system. An automatic monitoring system shall be provided to monitor and indicate continuously the performance and operating environment of the UPS. This monitoring system shall have the capability of indicating satisfactory operation and provide fault location or isolation within the UPS. Any malfunction of the monitoring system shall not impair the operation of the UPS. Parameters to be monitored shall be selected as follows:

(a) Most indicative of total UPS performance.
(b) Indicative of long-term potential of failure when abrupt changes or drifting trends occur.
(c) Indicative of imminent potential of impending faults or failure, or both with immediate indication of such fault including its location within.

3.4.5.1 Monitor panel. A monitor panel shall be provided with each UPS. The monitor panel shall utilize indicators (indicating lights or equivalent means) to show the operating conditions of each drawer or module. Associated with each operating drawer or module shall be threshold detectors that can be set at the high and low limits to be sensed. When these limits are exceeded, the indicator associated with the sensed point shall have an indication. The sense points shall be monitored on a continuous basis.

3.4.5.1.1 Local test points. Test points in accordance with MIL-STD-415 shall be provided on the monitor panel for each indicator. These test points shall be capable of being utilized during normal operation without affecting the UPS performance. The test points shall be compatible with the oscilloscope specified as test equipment (see 3.7.3.3).

3.4.5.1.2 External test points. An auxiliary receptacle shall be provided with parallel wires for each sensed input for the purpose of external measurement or monitoring. These test points for external monitoring shall be in accordance with MIL-STD-1326. The auxiliary receptacle shall be located on the top of the cabinet.

3.4.5.2 Control panel. A control panel shall be provided for the operation of the UPS. The control panel may be part of the monitor panel (see 3.4.5.1). The control panel shall have the following items:

(a) Operation switch or pushbuttons to operate the UPS.
(b) Selector switch for the by-pass switch (see 3.3.10) and the selector switch shall have positions "automatic", "test", and "by-pass".
(c) Unit-parallel operation selection switch which shall have "unit" and "parallel" positions.
(d) Control device to adjust the UPS output voltage plus or minus 5 percent.
(e) Indicating lights shall be provided to indicate:
    (1) When the battery is supplying power to the load(s), and provision for connection to a remote indicating light on a remote control panel.

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(2) When the load is being supplied directly from the input a.c. lines and by-passing the UPS. Provision shall be provided for connection of a remote indicating light on a remote control panel.
(3) UPS output overvoltage.
(4) UPS output undervoltage.
(5) UPS output overcurrent.
(6) Low battery voltage.
(7) Battery overvoltage.
(8) Low rectifier-charger output.
(9) Blower failure or insufficient cooling air.
(10) Blown inverter fuse.
(11) Blown rectifier-charger fuse.

(f) The color code of the indicating lights shall be as follows:

(1)	Red	Danger or emergency.
(2)	Green	Normal.
(3)	Amber	Abnormal.
(4)	Yellow	Open.
(5)	Clear, etched (white)	Power on or power available.

3.4.5.2.1 Mimic bus and status indicators. The control panel shall contain a mimic bus, which shall be arranged in a power flow diagrammatic configuration and shall show by indicator lamps the open/closed status of contacts plus the on/off status of the inverter and rectifier-charger. The mimic bus indicators and their associated operating controls in 3.4.5.2 shall be integrated to the maximum extent possible. Indicator lamps shall be replaceable from the front of the panel without affecting the operation of the UPS.

3.4.5.2.2 Control panel metering. Selector switches and instruments shall be provided on the control panel to indicate the parameters listed below:

(a) A.c. output voltage.
(b) Battery voltage.
(c) Battery current (d.c. ammeter to be center scale suitably marked for reading battery charge and discharge current).
(d) Elapsed time meter which shall have a minimum read-out of 99999 hours and shall conform to MIL-M-7793.
(e) A.c. output current.

3.4.5.3 Memory indicator. Memory indicator shall be provided on each drawer to show the malfunction of the drawer. The correct indication shall be retained on the drawer after the UPS input power has been turned off or the drawer has been removed from the UPS. An electrical reset switch shall be provided on the drawer. The memory indicator shall only be reset upon reapplication of the UPS input power.

3.4.6 Identification of UPS divisions by reference designations. To provide identification that will aid documentation and facilitate maintenance, each part, module, drawer, and cabinet of the UPS shall be identified by reference designations in accordance with ANSI Y32.16 and with 3.4.6.1 and 3.4.6.2.

3.4.6.1 Identification of parts leads. For each polarized part, such as electrolytic capacitors and rectifiers, the polarity identification shall be marked on the part mounting surface in order to help assure proper replacement of the part. For multiple lead parts such as transistors, where proper circuit operation is dependent upon a specific lead hookup, appropriate lead identification shall be marked on the parts mounting surface to help assure proper replacement of the part.

3.4.6.2 Method of marking. Markings shall be permanent and legible. The markings on plastic or metallic materials shall be made by stamping, engraving, stenciling or rubber stamping with smudgeproof ink covered with a coat of clear lacquer, or silk screening. Decalcomanies or paper labels shall not be used.

3.4.6.3 Identification plate. The UPS shall be equipped with an identification plate. The identification plate shall be metallic of types A, B, C or H in accordance with MIL-P-15024 and MIL-P-15024/5.

3.4.6.4 Operation instruction plate. The UPS shall be equipped with an operation instruction plate. The plate shall conform to MIL-P-15024 and MIL-P-15024/5, types A, B, C or H (metallic only).

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3.4.7 Human engineering. Human engineering effort shall conform to MIL-H-46855.

3.4.8 Safety. The UPS shall be designed for the utmost safety of ship's personnel and equipment. Safety of the UPS shall conform to MIL-E-917. A safety statement shall be provided in accordance with the data ordering document (see 6.2.2). Personnel shall be protected from capacitor discharges in accordance with requirement 1 of MIL-STD-454. When the UPS input contactor (see 3.4.9) is in the off position, all circuits of the UPS shall be de-energized (only the input terminals of the input contactor and its associated controls and the batteries can be energized). The use of a solenoid which is allowed by requirement 1 of MIL-STD-454 shall be treated as non-standard parts as specified in 3.5.1.

3.4.8.1 Analyses. Analyses shall be performed to identify hazardous conditions for the purpose of their elimination or control (see 6.2.2). Analyses shall be made to examine the UPS, cabinets, drawers, and parts, and their interrelationship to include logistic support, training, maintenance and operational environments. The analyses shall be accomplished to do the following:

(a) Identify hazards and determine any needed corrective actions.
(b) Determine and evaluate safety considerations in tradeoff studies.
(c) Determine and evaluate appropriate safety design requirements.
(d) Determine and evaluate maintenance, operational, test, and logistic safety requirements.

3.4.8.1.1 The areas to be considered in the analyses shall include, but not be limited to the following:

(a) Utilize historical data for safety consideration.
(b) Isolation of energy sources.
(c) System environment constraints.
(d) Compatibility of materials.
(e) Safe operation and maintenance of the UPS.
(f) Training and certification pertaining to safe operation and maintenance of the UPS.
(g) Fire ignition and propagation sources and protection.
(h) Environmental factors such as equipment layout and lighting requirements and their safety implications.
(i) Fail safe design considerations.
(j) Human error analysis of operator functions, tasks, and requirements.

3.4.8.2 A summary of analyses performed and resolution of identified category I and II hazards, as defined in MIL-STD-882, shall be documented and reported to the command or agency concerned (see 6.2.2).

3.4.9 Input contactor. An input contactor shall be provided in the cabinet. The input contactor shall be activated from an operation switch mounted on the control panel (see 3.4.5.2). The contactor shall be in accordance with MIL-C-2212.

3.4.10 By-pass switch. A by-pass switch shall be provided in the cabinet. The by-pass switch shall be activated from a selector switch mounted on the control panel (see 3.4.5.2). The by-pass switch shall function as shown in the table in figure 1.

(a) When the selector switch is set at the "automatic" position, any malfunction or failure of the UPS shall automatically transfer the load(s) from the UPS to the input a.c. lines through the automatic transfer switch provided that the voltage and frequency of the input a.c. lines are within the steady-state limits of the type I power.
(b) When the selector switch is moved to the "by-pass" position, the load(s) shall be transferred from the UPS to the input a.c. line. The transfer shall be inhibited when the frequency and voltage of the input a.c. lines are outside the steady-state limits of the type I power.
(c) When the selector switch is set at the "test" position, the operation of the UPS, including the automatic transfer switch, can be tested under dummy load while the UPS load is by-passed to the input a.c. lines.

3.4.11 Battery circuit breaker. A circuit breaker shall be provided in the cabinet and shall be connected between the batteries and the rectifier-charger output as shown in figure 1. The battery circuit breaker shall be in accordance with MIL-C-17361 or MIL-C-17587.

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3.5 Component requirements.

3.5.1 Selection of parts. The UPS shall use parts, modules, and drawers. Parts, modules and drawers which perform similar functions shall be standardized and interchangeable to the maximum extent practicable. The number of different parts to be designed into the UPS shall be kept to a minimum consistent with its intended use. Selection of parts shall be in accordance with MIL-E-917. A parts control program shall be established in accordance with procedure I of MIL-STD-965.

3.5.2 Subcontracted parts. For the subcontracted parts which are acquired by the contractor for incorporation into the UPS and do not lose their separate identity when the UPS is completed, each of these parts shall be identified in the parts list and drawings (see 3.9.1) by the part manufacturer's name and part number in addition to the UPS contractor's part number.

3.5.3 Fuses. Fuses used in series with power semiconductors shall be of the type suitable for protection of semiconductor devices (see 3.3.7.4). Except for semiconductor protection, fuses shall be in accordance with MIL-F-15160. Voltage rating of fuses shall be not less than the maximum circuit application voltage.

3.5.4 Solid state circuitry. The UPS shall utilize all solid state circuitry, except the input contactor, by-pass switch, battery circuit breaker, and the batteries.

3.5.4.1 Power circuits. Power circuits shall be built as removable modules. Each module shall include a single main power thyristor or transistor with any complementary device that may be required and all the associated components (such as current limiting, voltage limiting, and heat dissipating) necessary to ensure proper and reliable operation of the thyristor or transistor in a practical circuit.

3.5.4.2 Thyristor or transistor firing circuit. Identical thyristor or transistor firing circuit modules shall be used throughout the UPS to the maximum extent possible. Firing circuits shall be designed for maximum noise immunity utilizing decoupling, shielding, low impedance, twisted pair cables, reverse bias, isolation, and proper wiring layout to maximum extent possible.

3.5.4.3 Electrical control circuits. The Standard Electronic Modules (SEM) in accordance with MIL-STD-1378 and MIL-M-28787 shall be used to the maximum extent possible for control circuitry. The UPS shall be designed to protect itself without damage to any parts, modules, or drawers in the event that any control circuit is shorted or opened. UPS performance is not required if a control circuit is shorted or opened.

3.5.5 Semiconductor devices. Semiconductor devices shall be chosen and applied in accordance with MIL-E-917 and as specified in 3.5.5.1 through 3.5.5.7. Unless otherwise noted, discrete semiconductor devices shall be silicon and level JANTX in accordance with MIL-S-19500.

3.5.5.1 Standardization promotion. When a justifiable need exists for a device not covered by an existing military specification, but which can be provided by extension of a series, or family, of military devices, such extension shall be employed in preference to choice of a device from a new family or series. The power semiconductor devices used (diodes, transistors, and thyristors) shall be presently available without further development. The power semiconductor shall have at least two sources of supply.

3.5.5.2 Device selection prohibition. The UPS shall meet all requirements without the need for device selection beyond that covered by the applicable device detail specification sheet of MIL-S-19500. Semiconductor devices of the same type designation shall be interchangeable without any restrictions, including sources of manufacture.

3.5.5.3 Reverse polarity prohibition. Devices of standard polarity shall be used wherever feasible. Devices of reverse polarity may be used only under the conditions applicable to the use of non-standard parts (see 3.5.1).

3.5.5.4 Stress factors and limits. Circuits shall be designed and devices shall be applied so that the stress limits specified in table IV shall not be exceeded under any condition of operation, standby, and during tests specified for the UPS. The values specified are limits and not design factors, nor design margins. The equipment designer shall determine and provide sufficient margin between the applied stresses and the specified limits to make sure that the required life and reliability are achieved. In no case shall the relative values of the steady-state voltage variation, over-voltage, transient voltage, and surge voltage spikes, for the circuit under consideration, be considered to be less

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than the values specified in 3.3.1, 3.3.1.1, and 3.3.1.2, except where the design includes features to limit or suppress over-voltage and surge voltage to lower values. When such exception is used, the contractor shall provide a detail description of the design features and how the circuit functions to limit peak over-voltage and surge voltage impressed on the semiconductor devices to whatever lower values apply. Circuits which employ thyristors or transistors shall be designed to suppress surge voltages at least to not more than three times the supply nominal rms voltage. Semiconductor devices shall be applied in accordance with 3.5.5.5. Surge voltage suppressors shall be in accordance with 3.5.5.6. Heat dissipators (heat sinks, cooling fins) shall be in accordance with 3.5.5.7.

TABLE IV. Application stress limits and rerating factors.

Stress factor	Stress factor limit in relation to rating factor		Rating factor
	Lower	Upper	
Semiconductor devices:			
(a) General:			
(1) Ambient temperature design:			
a. Minimum T_{AL}	T_{AD}-50°C	T_{AD}-30°C	T_{AD} = equipment design ambient
b. Maximum T_{AH}	T_{AD}+15°C	T_{AD}+35°C	T_{AL} = design ambient temperature lower limit
(2) Ambient temperature measured:			
a. Minimum, start-up	T_{AL}		
b. Maximum, operating	---	T_{AH}	T_{AH} = design ambient temperature higher limit
(b) Rectifiers and thyristors:			
(1) Nominal working peak reverse voltage	---	0.5	Repetitive peak reverse voltage
(2) Repetitive peak reverse voltage	---	.7	Repetitive peak reverse voltage
(3) Non-repetitive peak reverse voltage	---	1.0	Repetitive peak reverse voltage, or forward blocking voltage (thyristors)
(4) D.c. blocking voltage	---	0.4	Repetitive peak reverse voltage
(5) Nominal working peak forward voltage (thyristors)	---	.5	Forward blocking voltage
(6) Repetitive peak forward voltage (thyristors)	---	.7	Forward blocking voltage
(7) Non-repetitive peak forward voltage (thyristors):			
a. Self-firing unacceptable	---	.8	Forward blocking voltage
b. Occasional self-firing acceptable	---	1.0	Forward blocking voltage
c. Same as b. and di/dt = 0.3 maximum allowable	---	1.0	Peak forward blocking voltage
(8) Reapplied forward blocking voltage (thyristors)	---	0.7	Forward blocking voltage at rated dv/dt, junction temperature and turn-off time

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TABLE IV. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation to rating factor		Rating factor
	Lower	Upper	
Semiconductor devices:			
(b) Rectifiers and thyristors (continued):			
(9) Minimum duration of voltage reverse (thyristors)	2.0	---	Turn-off time
(10) Maximum rate of rise of forward blocking voltage (thyristors)	---	0.5	Minimum rate of rise, forward blocking voltage (dv/dt)
(11) Average forward current:			
a. Nominal value	---	.7	Forward current derated for conductive angle, wave form stud or case temperature T_C plus 15°C (ambient temperature T_A plus 15°C for lead-mounted devices) and duty cycle
b. Maximum value	---	1.0	
(12) Repetitive peak forward current:			
a. Nominal value	---	0.7	Repetitive peak forward current at maximum case temperature (maximum ambient temperature for lead-mounted devices)
b. Maximum value	---	1.0	
(13) Surge current:			
a. Peak, one cycle, non-repetitive	---	0.7	Surge current, peak, half cycle (1/120 sec.)
b. Effective peak and cycles duration	---	$0.2+0.28 \log \frac{f}{n}$	non-repetitive (log = logarithm to base 10, f = pulse frequency in Hz n = number of pulses, t = time in milliseconds
c. Effective amplitude and duration sub-cycle	---	$1. - 0.6t$	
(14) Peak current during turn-on (thyristors)	---	0.7	Peak current during turn-on
(15) Maximum rate of current rise during turn-on (thyristors)	---	.5	Maximum rate of current rise during turn-on
(16) Peak forward gate current (thyristors):			
a. Maximum (for firing)	---	.7	Peak forward gate current
b. Minimum (for firing)	2.0	---	Forward gate current, minimum to cause firing
c. Maximum (during blocking)	---	0	Forward gate current, minimum to cause firing
(17) Peak forward gate voltage (thyristors):			
a. Maximum (for firing)	---	0.7	Peak forward gate voltage
b. Minimum (for firing)	2.0	---	Minimum firing voltage
c. Maximum (during blocking)	---	0	Minimum firing voltage
(18) Peak reverse gate voltage maximum (thyristors)	---	0.7	Peak reverse gate voltage

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TABLE IV. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation to rating factor		Rating factor
	Lower	Upper	
Semiconductor devices:			
(b) Rectifiers and thyristors (continued):			
(19) Maximum gate power dissipation (thyristors):			
a. Average	---	0.7	Average gate power dissipation
b. Peak	---	1.0	Peak gate power dissipation
(20) Maximum junction temperature (calculated):			
a. Average[1/]	---	T_j (maximum)-40°C	T_j (maximum) = maximum junction temperature
b. Peak[2/3/]	---	T_j (maximum)-25°C	
(c) Voltage reference diodes:			
(1) Bias current	0.8	1.2	Zener test current (value for rated Zener voltage)
(2) Diode reverse current (Zener current):			
a. Minimum	.1	---	Zener test current (value for rated Zener voltage)
b. Maximum	---	0.5	Maximum Zener current
c. Maximum, continuous duty	---	.3	Maximum Zener current
(3) Junction temperature (calculated)	---	100°C	
(d) Voltage regulator diodes:			
(1) Diode reverse current (Zener current):			
a. Minimum	0.1	---	Zener test current (value for rated Zener voltage)
b. Maximum (steady-state)	---	0.5	Maximum Zener current
c. Maximum, continuous duty	---	.3	Maximum Zener current
d. Surge	---	.5	Surge current
(2) Junction temperature (calculated)			
a. Average (steady-state)[1/]	---	T_j (maximum)-50°C	T_j (maximum) = maximum junction temperature
b. Peak[2/3/]	---	T_j (maximum)-25°C	
(e) Transistors:			
(1) Collector to emitter voltage	---	0.5	Collector to emitter voltage, VCEO
(2) Collector current	---	.5	Maximum collector current
(3) Collector power dissipation:			
a. Maximum (steady-state average)	---	.5	Collector power dissipation, Pc, derated for $T_C + \Delta T$ (or $T_A + \Delta T$) where ΔT = 50°C for silicon or 15°C for germanium
b. Peak	---	.7	

See footnotes at end of table.

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TABLE IV. Application stress limits and rerating factors. - Continued

Stress factor	Stress factor limit in relation to rating factor		Rating factor
	Lower	Upper	
Semiconductor devices: (e) Transistors (continued): (4) Junction temperature (calculated) a. Average (steady-state)[1/]	---	T_j (maximum)-50°C	T_j (maximum) = maximum junction temperature
b. Peak[2/3/]	---	T_j (maximum)-25°C	

[1/] Based on worst case conditions including maximum value of device thermal resistance, least effective cooling, and most severe duty cycle.

[2/] Based on worst case conditions including instantaneous power dissipation and transient thermal impedance. This refers to peak valve as a function of time and not junction hot spot.

[3/] Applicable for devices capable of 5 watts or greater power dissipations at 25°C case temperature and where duty cycle involves varying load levels of shorter duration than 3 times the thermal time constant of the device.

3.5.5.5 Application limitations. The following limitations and design guides shall apply in the application of semiconductor devices:

(a) Devices designed and rated for conduction cooling shall be mounted on metallic heat dissipators. Due regard shall be given to the compatibility of materials and to the finish coatings on parts mounted together in an assembly. Particular attention shalll be given to the design of connections and joints in both electrical and thermal circuits to make sure that their effectiveness will be maintained for the life of the UPS under adverse shipboard conditions, including shocks, vibration, thermal cycling, high and low humidity, and salt-laden atmosphere. Means for maintaining positive contact pressure on bolts or clamped joints, under all service conditions, taking into account the aging (including creep, cold flow, relaxation, fatigue, shrinkage, and so forth) of materials, shall be included.

(b) Direct mounting of devices to their heat dissipator with electrical insulation of the heat dissipators, as necessary, is preferred to the use of intervening electrical insulating material between the devices and their heat dissipators. This preferred arrangement shall be used except where it is impracticable to insulate the heat dissipators, or where other technical considerations make electrical insulation of the semiconductor device from its mounting surface necessary.

(c) Semiconductor devices shall not be operated in parallel to provide current capacity.

(d) Semiconductor devices (diodes, rectifiers, thyristors, and transistors) shall not be operated in series.

3.5.5.6 Surge voltage suppression. Circuits containing devices which do not have inherent capability of withstanding voltage surge of the level specified in 3.5.5.4 without exceeding allowable stresses shall be protected by incorporating adequate surge voltage suppression within the circuit. The internal generation of voltage surges shall be minimized with respect to both frequency of occurrence and amplitude. Means shall be provided within the circuit both for suppressing any remaining voltage surges generated within the circuit that would otherwise be excessive and for preventing external propagation at a peak level greater than three times the rms voltage for the a.c. output circuit and that permitted by 3.3.1 and 3.3.1.1 for the input circuit. Surge voltage suppressors shall be in accordance with MIL-E-917.

3.5.5.7 Heat dissipators. Heat dissipators for semiconductor devices shall be constructed of metallic material which is either corrosion-resistant, or which is treated or coated to resist corrosion under shipboard environmental conditions. Surfaces which contact the semiconductor device and surfaces to which electrical contacts are made shall not be painted or anodized, but shall be plated or otherwise coated to form surfaces to which the connections or junctions are made and which will retain their mechanical, thermal and electrical effectiveness for the life of the equipment. The method of processing and

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the preparation of the connections and joints shall be as specified on the applicable heat dissipator drawing. Heat dissipators shall be sized to dissipate the heat generated by the semiconductor device of highest losses that may be expected to conform to the applicable semiconductor device detail specification sheet of MIL-S-19500, and to do so under the highest ambient temperature conditions without exceeding the limiting temperatures specified in table IV. Maximum power dissipation may be estimated from calculations taking into account the mode of operation of the device in the circuit, the worst stress levels applied, and the limiting device characteristics as covered by the applicable semiconductor device detail specification sheet of MIL-S-19500. Maximum power dissipation may also be estimated by using published data for a particular brand of device, and scaling up the losses using a factor calculated from parameter limits applicable to the particular brand of device and specification limits, taking into account the influence of the parameters considered on the losses under the mode of operation in the circuit and allowing for any difference in form factor involved in the way the values are given. Where maximum power dissipation at rated conditions is specified and controlled by the applicable semiconductor device detail specification sheet of MIL-S-19500, this value, corrected for the stress levels of the application, shall be used in the design.

3.5.6 Indicating light. Indicating lights shall conform to MIL-L-3661 and shall have push-to test provision. The use of light-emitting-diodes (LED) as indications in monitoring system is acceptable.

3.5.7 Electrical parts derating. Electrical parts, other than the semiconductors, shall be derated in accordance with NAVSEA 0967-LP-597-1010. Circuits shall be designed and parts shall be applied so that the stress levels limits specified in NAVSEA 0967-LP-597-1010 will not be exceeded under all operating conditions specified herein.

3.6 Environmental requirements.

3.6.1 Shock. The UPS shall withstand the high-impact, type A, grade A, class I shock test of MIL-S-901. Resilient mounts shall not be used for any parts, modules and drawers in order to meet the shock test requirement. Resilient mounts may be used to reduce the airborne and structureborne noise levels of the UPS. If resilient mounts are used for noise attenuation, they shall be shock tested with the UPS.

3.6.2 Vibration. The UPS shall be designed and tested to meet the type I vibration tests of MIL-STD-167-1 up to and including 50 Hz. Cabinets and the battery rack enclosure(s) of the UPS shall withstand the type I vibration test of MIL-STD-167-1 without any sway braces.

3.6.3 Noise. The UPS shall be solidly mounted to the ship structure and shall be designed and tested to meet the requirements of MIL-STD-740, except that the airborne noise requirements shall be as specified in table V.

TABLE V. Acceptable octave band sound pressure levels in (dB re 20 µPa).

Frequency in Hz	31.5	63	125	250	500	1K	2K	4K	8K
Sound pressure level	95	90	85	82	79	76	75	75	75

3.6.4 Electromagnetic interference emission and susceptibility. The UPS shall conform to the electromagnetic interference limit in accordance with MIL-STD-461. The UPS shall conform to the EMI limits of class IIA equipment except that the narrowband conducted emission levels of CE01 and CE03 as specified in figures 7 and 8 shall be used in place of the broadband limits CE01 and CE03 of MIL-STD-461. Cable and enclosure bonding and grounding shall conform to MIL-STD-1310. The UPS shall be designed to minimize susceptibility of its control circuits to EMI in accordance with MIL-STD-461.

3.6.5 Ambient temperature. The UPS shall start and operate satisfactorily and conform to all requirements specified herein throughout an ambient temperature range of 0°C to 50°C.

3.6.6 Cooling. The UPS shall be cooled by ambient air of a maximum temperature of 50°C. The cabinets may be forced-air cooled. Air filter conforming to MIL-F-16552 and Drawing S3801-1170895 shall be used with forced-air cooling. The battery rack enclosure(s) shall be convection cooled. Protection of the UPS against cooling fan failure shall be provided by means of automatic transfer of the load(s) from the UPS to the input a.c. lines.

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3.6.7 Accessibility. Front access shall be provided to all parts, modules and drawers of the UPS. Side or back access shall not be required for operation and maintenance.

3.6.8 Ship motion. The UPS shall operate satisfactorily when inclined 15 degrees permanently in any direction and when installed on board ships having the following pitch and roll:

	Magnitude	Period
Pitch	10 degrees	5 seconds
Roll	45 degrees	8 seconds

3.6.9 Lifting provision. Each cabinet shall have removable eyebolts provided on top to permit hoisting and handling. The eye-bolts shall be located to balance the weight of the cabinet as prepared for shipping in the upright position. The lifting attachment shall be of sufficient capacity to carry at least five times the total weight of the cabinet. The cabinet shall remain dripproof when the eyebolts are removed.

3.6.10 Mounting. The UPS shall be designed for rigid deck mounting. Mounting provisions shall be arranged for easy access and adjustment of mounting screws or bolts without disassembly of the UPS or the use of special tools (see 3.8.3). The deck-mounted UPS shall have mounting feet or runners of sufficient elevation for easy installation. Unless the enclosure is designed to carry the weight of parts and subassemblies enclosed, the UPS shall be mounted in such a manner that the weight is transmitted directly to the ship structure and not borne by the wall of the enclosure.

3.6.11 Humidity conditions. The UPS shall be designed to operate with relative humidities ranging up to 95 + 5 percent for both intermittent and continuous periods including conditions wherein condensation takes place in and on the UPS cabinets in the form of both water and frost.

3.6.12 Clearance for ventilation and accessibility. Maximum clearance allowed between the UPS cabinet(s) and adjacent equipment or structure shall be as follows:

(a) Bottom . . . . . No clearance allowed
(b) Sides . . . . . No clearance allowed
(c) Rear . . . . . 4 inches
(d) Top . . . . . 5 inches
(e) Front . . . . . 30 inches

3.7 Reliability and maintainability requirements.

3.7.1 Definitions. Definitions of terms for reliability and maintainability shall be in accordance with MIL-STD-721, unless otherwise specified herein.

3.7.2 Reliability. The reliability measure shall be Mean-Time-Betwee-Failures (MTBF). The values of MTBF shall be as follows in accordance with MIL-STD-781 (see 4.7.43):

(a) Upper test MTBF (θ_0) - 4000 hours
(b) Lower test MTBF (θ_1) - 1334 hours

3.7.2.1 Definition of failure. The definition of failure for the purpose of determining the achievement of reliability characteristics shall be:

(a) The input and output performance not in accordance with requirements specified herein.
(b) Malfunction requiring transfer and shutdown because of damage or incipient damage.

3.7.3 Maintenance concept. The principal objective of maintenance design shall be to provide for the minimum time of interruption for the purpose of maintenance. Corrective maintenance shall be accomplished at three levels as follows:

(a) UPS at site of installation. Removal and replacement of entire drawers, employing only the monitoring system to detect and identify the faulty drawers.

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(b) Drawers after removal from the cabinet. Removal and replacement of circuit boards, modules, and parts, employing the test equipment specified herein.
(c) Printed circuit boards, modules, and parts after removal from the drawers. Repair of repairable circuit boards and modules employing test equipment specified herein.

3.7.3.1 Corrective maintenance. The times for corrective maintenance shall be in accordance with the indices and values below:

(a) Substitution of drawers. For the substitution of drawers, including identification, removal, and replacement of faulty drawers, and checkout after replacement, the times as defined in MIL-STD-471, method 2, shall be:
(1) Specified Mean-Corrective-Maintenance-Time (M_{ct}): not more than 10 minutes.
(2) Specified Maximum-Corrective-Maintenance-Time (M_{maxct}): not more than 25 minutes.
(b) Substitution of circuit boards and parts within drawers. For the substitution of circuit boards and parts within drawers, the cooling fan(s), the input contactor, the EMI filters and the power transformers, the times, as defined in MIL-STD-471, shall be (test for mean maintenance time):
(1) Specified M_{ct}: not more than 2 hours.
(2) Specified M_{maxct}: not more than 5 hours.
(c) Repair of repairable circuit boards and modules after removal from drawers. For the repair of circuit boards and modules by substitution of parts, the time, as defined in MIL-STD-471, method 3, shall be:
(1) Equipment Repair Time: not more than 68 minutes.

3.7.3.2 Maintainability design. The UPS shall conform to NAVSEA 0967-LP-312-8010.

3.7.3.3 Test equipment. The UPS shall require the use of only the following test equipment for maintenance and repair:

(a) Clamp-on volt ammeter, 0 to 600 amperes and 0 to 600 volts, a.c., National Stock Number (NSN) 1N6625-284-0855 or 1N6625-879-1231; Nomenclature: CV-633 or AN/USM-33.
(b) Multimeter, 0 to 5000 volts, d.c.; 0 to 1000 volts, a.c.; 0 to 10 amperes, d.c.; and 0 to 30 megohm range; Nomenclature: AN/PSM-4.
(c) Oscilloscope, X-Y, low frequency: Nomenclature; AN/USM-368 (Hewelett Packard Model 130C or equivalent).
(d) D.c. milliammeter, 0 to 150 milliamperes range; NSN: 2N6625-945-2759; Nomenclature: CV-931-4904003.
(e) A.c. voltmeter, 0 to 600 volts range; NSN: 1N6625-184-5337; type MR71W600ADVV in accordance with MIL-M-16034 and MIL-M-16034/9.

3.7.4 Reliability and maintainability program and program plans. The contractor shall conduct reliability and maintainability programs as specified herein. The programs shall be separate but coordinated. The contractor shall conduct the programs in accordance with the NAVSEC accepted program plans.

3.7.4.1 Reliability program plan. The reliability program plan shall be prepared in accordance with the data ordering document (see 6.2.2) and shall provide for the formal application of the reliability management procedures and reliability design procedures of MIL-STD-785 specified herein.

3.7.4.1.1 Reliability management procedures. The reliability management procedures shall include the following elements of the detailed requirements of MIL-STD-785:

(a) Reliability organization.
(b) Management and control.
(c) Subcontractor and contractor reliability programs.
(d) Program review.
(e) Failure data collection, analysis, and corrective action. Provision shall be made by the contractor for complete reporting on every malfunction, its diagnosis, and the corrective action taken (design modification, etc.). The report (see 6.2.2), shall cover check-out and testing performed by the contractor on the breadboard and prototype units and testing performed by the Navy personnel during technical evaluations in Naval laboratories and operation evaluation on board Naval ships.
(f) Failure summaries.
(g) Status reports.

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3.7.4.1.2 Reliability design procedures. The contractor shall provide for the attainment of the reliability design in the UPS by applying the following elements of the detailed requirements of MIL-STD-785 for the design, the production, and the logistics planning:

(a) Design techniques (see 3.7.2.2).
(b) Reliability analysis (see 3.7.4.1.3).
(c) Parts reliability.
(d) Failure modes, effects, and criticality analysis (see 3.7.4.1.4).
(e) Reliability critical items.
(f) Effects of storage, shelf-life, transportation, handling, and maintenance.

3.7.4.1.3 Reliability analysis. The reliability analysis shall be prepared in accordance with the data ordering document (see 6.2.2) and shall comply with part stress analysis prediction method of MIL-HDBK-217. Failure rates shall be selected from MIL-HDBK-217. When component failure rates are not available in MIL-HDBK-217, failure rates from other sources shall be used. Failure rates from other sources shall be fully documented and substantiated. For reliability analysis, the failure rate of each component shall be calculated based on the maximum ambient temperature in which the component operates in the UPS. The ambient temperature at each component shall be calculated based on the maximum input cooling air temperature of 50°C. When using MIL-HDBK-217, the environmental factor shall be Naval unsheltered.

3.7.4.1.4 Failure modes, effects, and analysis (FMEA). The FMEA shall be prepared in accordance with the data ordering document (see 6.2.2).

3.7.4.2 Maintainability program plan. The maintainability program plan shall be prepared in accordance with the data ordering document (see 6.2.2) and shall provide for the formal application of the maintainability management procedures and maintainability design procedures specified herein.

3.7.4.2.1 Maintainability management procedures. The maintainability management procedures shall include the following elements of the detailed requirements of MIL-STD-470:

(a) Prepare maintainability program plan (see 6.2.2).
(b) Incorporate and enforce maintainability requirement in subcontractor and vendor specification.
(c) Integrate other items.
(d) Participate in design reviews.
(e) Establish data collection, analysis, and corrective action system.
(f) Prepare maintainability status reports.
(g) Perform maintainability demonstration.

3.7.4.2.2 Maintainability design procedures. The contractor shall provide for the attainment of maintainability design in the UPS by applying the following elements of the detailed requirements of MIL-STD-470 for design, for production, and for logistics planning:

(a) Perform maintainability analysis.
(b) Prepare inputs to the detailed maintenance concept and detailed maintenance plan.
(c) Perform design trade-offs.
(d) Predict maintainability parameter values.

3.7.4.2.3 Maintenance engineering analysis (MEA). The contractor shall prepare a MEA in accordance with the data ordering document (see 6.2.2). The analysis shall provide the basis for assurance that the design of the UPS is consistent with the maintenance concept (see 3.7.3) and with the reliability and maintainability indices (see 3.7.2 and 3.7.3) herein. The MEA shall result in an estimate of the maintenance requirements and tasks for their accomplishment, number of persons, involved and their skills and tools, equipment and facilities required to maintain the UPS when installed on board ships. It shall cover both organizational level and intermediate level maintenance. The maintenance burden for each level shall be separately indicated. The MEA shall apply, but not be limited, to the following:

(a) Corrective maintenance.
(b) Planned maintenance of mechanical components.
(c) Reconditioning or repair of replaceable modules.
(d) Selection of repair or throw-away for particular modules.
(e) Minimization of repair-induced faults.

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3.7.4.2.3.1 MEA documentation. The final report of MEA shall reflect the design of the UPS in condition for delivery. The report of MEA shall be in narrative form supplemented by the completed work sheets (numbers I through VII of MIL-M-24365).

3.7.5 Reliability and maintainability final report. The contractor shall prepare a report in accordance with the data ordering document (see 6.2.2) of the reliability and maintainability of the UPS in configuration for delivery, supported and documented by the results of the reliability management, reliability design, maintainability management, maintainability design procedures, and performance tests accomplished under the contract. The report shall include, but not be limited to, the following items:

(a) Assessment of MTBF and Mean-Time-To-Repair (MTTR) achieved.
(b) Definitive evidence of conformity of the design with the maintenance concept (see 3.7.3).
(c) Estimate of potential reliability and maintainability in operational use.
(d) Summaries of mathematical models, block diagrams, failure modes and effects analysis, allocations and trade-offs.
(e) Rationale for significant trade-offs.
(f) Failure summaries.
(g) Conclusions, identification of potential problem areas, and recommendations for action with respect to items of relatively high criticality in the UPS.
(h) Results of failure data collection analysis and corrective action procedures.

The report shall include the supporting information for the reliability and maintainability data specified for inclusion in the engineering drawings.

3.8 Repair parts.

3.8.1 Interchangeability. Each repair part shall be equal to the original and suitable for replacement without special tools, and without the removal and replacement of rivets or pins dependent upon a force fit.

3.8.2 Provisioning. Separate provisioning technical documentation shall be prepared and submitted in accordance with MIL-STD-1375 for the equipment. When interim repair parts are acquired with the equipment (see 6.2.1), the range and quantity of interim repair parts shall be in accordance with MIL-STD-1375.

3.8.3 Special tools. For purposes of this specification, test equipment or tools not listed in the Federal Supply Catalog or in the ship's consolidated shipboard allowance list is a special tool. The need for special tools shall be minimized, and such need shall be specifically identified. The basis for choosing a design which makes necessary the use of a special tool other than those specified or required herein (see 3.7.3.3) shall be justified prior to incorporating the feature into the UPS design. The special tool may be used only under the conditions applicable to the use of non-standard parts (see 3.5.1). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service Management Area (DCASMA)).

3.9 Technical data. The contractor shall prepare the following technical data in accordance with the data ordering documents included in the contract or order (see 6.2.2), and as specified in 3.9.1 through 3.9.2.

3.9.1 Drawings. Drawings shall be prepared in accordance with the data ordering document (see 6.2.2). Drawings under the same type (see 3.9.1.1 through 3.9.1.4) shall be incorporated into a single set of drawings. Sheets of the drawings in a set shall be of the same size. The minimum size of the drawings shall be 17 by 22 inches and the maximum size shall be 28 by 40 inches.

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3.9.1.1 Design evaluation drawings. Design evaluation drawings shall be provided in accordance with the data ordering document (see 6.2.2). Design evaluation drawings shall contain the following data:

(a) Schematic diagrams showing the circuits, parts, instrumentation, and protective devices. Each part and subdivision of the equipment shall be identified by its reference designation.

(b) List of material presenting the following data in tabular form. (List of material shall exclude ordinary hardware items such as fasteners subassembly or part brackets, and other similar hardware items. Hardware items, materials and finishes employed in the mounting of semiconductor devices shall be identified and listed. Heat sink and enclosure assemblies shall be identified and listed. Hookup wire type shall be indicated. Number of each part required.):
  (1) Piece number (for flags of outline drawing).
  (2) Name of part (for example: transformer, resistor diode, shaft).
  (3) Applicable reference designation for electrical parts (for example: T3, R16, CR5).
  (4) Military or Federal specification number, MS number, or NAVSEA drawing number, if any.
  (5) Type, class, grade, size or other classification of any referenced specification or MS sheet.
  (6) Part number or identification assigned by the manufacturer.
  (7) Name of manufacturer of part.
  (8) Part number or identification assigned by part manufacturer.
  (9) National stock number, if known.
  (10) Rating of part from part manufacturer's catalog (for example: resistor 10-ohm, 25-watt; diode - 200 PRV, 750 milliamps, and similar ratings).
  (11) Application or stress level on part as used under rated load condition. (For example: resistor 7.1 watts; diode - 91 PRV, 400 milliamperes, and other similar applications).

(c) Transformer and wound component data:
  (1) Core data:
    a. Lamination form - EI, toroid, or other form.
    b. Lamination material and gage.
    c. Stack and interleave dimensions.
    d. Construction - open, tape wound cut core, and other construction.
    e. Core box.
    f. Ground insulation.
  (2) Winding data:
    a. Insulation class - class B, H, or other.
    b. Winding identification - same terminal designation as given in schematic and connection diagrams.
    c. Wire size and insulation type, for example, AWG 10, type B2.
    d. Number of turns.
    e. Voltage.
    f. D.c. resistance at 25°C.
    g. Material of insulation used between winding layers and between windings.
  (3) Varnish impregnation:
    a. Varnish type and specification.
    b. Varnish manufacturer and material identification.
    c. Impregnation procedure - preheat temperature and time, number of dips and bakes, temperature and time of bakes, etc.
  (4) Encapsulation:
    a. Potting material - type and specification.
    b. Potting material manufacturer and identification.
    c. Potting procedure - resin mixing ratio, coring temperature and time, etc.

(d) Semiconductor devices worst case application data including the following listed items plus complete identification for ordering from potential contractors (including JEDEC number, where applicable):
  (1) Rectifiers using silicon diodes:
    a. Reference designations and piece number.
    b. Maximum reverse voltage peak and rms (complete rectifier circuit).

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c. Maximum load current d.c. amperes (complete circuit).
d. Maximum diode case temperature at design ambient - use for diodes rated against case temperature.
e. Maximum ambient temperature directly adjacent to diodes - use for diodes rated against ambient temperature - no cooling plates.
f. Heat sink arrangement and effective cooling area per device (or cooling plate size, thickness and material) cooling media and velocity.
g. Circuit (for example, single phase bridge, etc.).
h. Total diodes in circuit.
i. Diodes in series per arm.
j. Diodes in parallel per arm.
k. Mounting details.

(2) Voltage regulator diodes:
a. Reference designations and piece number.
b. Range of actual voltage levels which circuit will accept (with or without circuit adjustment) from regulator diodes and still give required equipment performance: _____ volts to _____ volts.
c. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum) plus or minus _____ millivolts.
d. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds), expressed in milliamperes.
e. Maximum transient reverse current under any operating condition expressed in _____ milliamperes for _____ seconds duration and frequency of occurrence.
f. Maximum ambient temperature at diode (no cooling plate).
g. Maximum case temperature of diode (complete for diode rated against case temperature).
h. Heat sink arrangement and effective cooling area per diode (or cooling plate size, thickness and material) cooling media and velocity.
i. Total diodes in circuit.
j. Diodes in series per circuit element (complete where more than one circuit element is used in the voltage regulator such as in a bridge type regulator circuit). (Operation in parallel is not permitted).
k. Mounting details.

(3) Voltage reference diodes (see regulator diodes also):
a. Reference designations and piece number.
b. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): _____ volts to _____ volts.
c. Error permitted in an installed voltage reference diode under all conditions of operation: plus or minus _____ millivolts.
d. Circuit design current through voltage reference:
1. Nominal or average value: (expressed in milliamperes).
2. Regulation about nominal value: (expressed in milliamperes above and below nominal).
e. Maximum ambient temperature (include internal heat rise above outside design ambient): (expressed in _____°C.).
f. Number of voltage reference elements in series: (operation in parallel not permitted).

(4) Transistors:
a. Reference designations and piece number.
b. Maximum collector to emitter voltage.
c. Maximum collector to base voltage.
d. Maximum collector current.
e. Maximum base current.
f. Maximum collector power dissipation.
g. Maximum ambient temperature (include internal heat rise at transistor at design ambient) in °C.

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h. Maximum case temperature of transistor at equipment design ambient and any condition of required equipment operation °C. (Where rating is based on case temperature).
i. Heat sink arrangement and effective cooling area per transistor (or cooling plate size, thickness and material) cooling media and velocity.
j. Mounting details.

(5) Silicon controlled rectifiers:
a. Reference designations and piece number.
b. Nominal working peak forward and reverse voltages.
c. Repetitive peak forward and reverse voltages (maximum).
d. Nonrepetitive peak forward and reverse voltages (maximum).
e. False firing, or self-firing, restrictions.
f. Repetitive peak forward current (nominal and maximum values).
g. Surge current (peak for nonrepetitive value occurring within the conduction portion of a single cycle but after turn-on has occurred, effective amplitude and cycles duration for multicycle surges, or amplitude and duration for sub-cycle pulse duration).
h. Peak current and maximum rate of current rise during turn-on.
i. Minimum time from the instant of voltage reversal to the instant when voltage is re-applied in the forward direction.
j. Peak forward gate current (minimum and maximum, per device).
k. Peak forward gate voltage (minimum and maximum, per device).
l. Peak reverse gate voltage (maximum, per device).
m. Maximum gate power dissipation (average and peak, per device).
n. Maximum device case temperature at equipment design ambient.
o. Heat sink arrangement and effective cooling area per device, (or cooling plate size, thickness and material).
p. Cooling medium, its temperature and velocity.
q. Circuit.
r. Total devices in circuit.
s. Devices in series per arm.
t. Devices in parallel per arm.
u. Mounting details.

(6) Selenium surge suppressors:
a. Reference designations and piece number.
b. Steady-state working voltage (a.c. runs or d.c. average, as applicable).
c. Sustained overvoltages (a.c. runs or d.c. average, as applicable; 30 seconds, or longer, duration).
d. Transient voltage and duration (a.c. repetitive peak or d.c. effective square wave for overvoltage duration less than 30 seconds).
e. Unsuppressed value of the potentially available nonrepetitive peak surge voltage.
f. Rate of voltage rise.
g. Suppression voltage (suppressed value of the surge voltage).
h. Nonrepetitive surge current.
i. Power dissipation during surge and effective duration.
j. Maximum and minimum operating temperatures.
k. Maximum and minimum ambient temperatures at the suppressor.
l. Mounting and connection arrangement.
m. Cooling medium, its temperature and velocity.
n. Identify the suppressor manufacturer's technical data bulletin applicable to each product certified for the application. For each application indicate which product is installed in the equipment and which is the alternate product that may be used for replacement purpose, when desired.
o. Provide the following product rating data:
1. Stack steady-state voltage rating (a.c. rms or d.c. average, as applicable).
2. Stack clamping voltages (new and aged).
3. Peak discharge current (corresponding to clamping voltages).
4. Polarization (polarized or nonpolarized).

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5. Maximum energy (watt-seconds) at rated clamping voltage (aged) for single pulse (nonrepetitive) current.
6. Maximum pulse duration (microseconds) for repetitive pulses at supply frequency (specify cycles per second) and amplitude equal to clamping voltage (aged).
7. Maximum ambient temperature at which rated values in o., 1 through 6 apply.
8. Maximum operating temperature.

p. Provide product identification information, including the following:
1. Manufacturer's name and plant address.
2. Manufacturer's type and catalog numbers.
3. Government specification and type designation, if applicable.
4. Construction, outline and mounting dimensions.
5. Circuit.
6. Terminal arrangement, marking, and interconnections.
7. Spacing between cells.
8. Cell size.
9. Total cells per stack.
10. Cells in series per circuit element.
11. Cells in parallel per circuit element.
12. Identification plate information.
13. Stack weight.
14. Protective finish.
15. Identification of mounting and connection hardware provided.

(e) Mechanical packaging data:
(1) Cabinet structure:
a. Construction details and dimensions.
b. Types of welding, brazing, or bolting.
c. Structural material and specifications.
d. Side panel material and specifications.
e. Corrosion resistance treatment.
f. Types of fasteners.
(2) Removable modules and drawers:
a. Construction details and dimensions.
b. Weight of each module and drawers.
c. Means of securing and alinement.
d. Parts arrangement.
e. Details of receptacle and electrical connections.
f. Electromagnetic interference provisions.
g. Cooling air flow direction.
h. Hardware corrosion treatment.
(3) Miscellaneous units or parts:
a. Power cable, conductors and busses - arrangement, supporting and cooling.
b. Cable harness grouping, routing, and protection.
c. Sizes of cables, conductors, and hook-up wire and insulation classes.

(f) Reliability and maintainability data:
(1) Parts accessibility.
(2) Designate repairable, replaceable, and throw-away parts or modules.
(3) Integrated circuits and the applicable MTBF.
(4) Design guidelines showing allowable stress level for each type of part or unit.
(5) Degree of redundancy.
(6) Fault isolation method.
(7) Monitoring system and test points.
(8) On-line repair consideration.
(9) Type of thyristor firing circuit and retention of noise immunity.
(10) Type of control circuit amplifiers.

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3.9.1.2 Interface control drawings. Interface control drawings shall be in accordance with data ordering document (see 6.2.2). Interface control drawings shall contain the following data:

(a) Outline drawing showing front, side, top and section view as necessary to present information as follows:
  (1) Overall mounting dimension.
  (2) Maximum weight.
  (3) Location and size of mounting hole and ground stud.
  (4) Cable entrance provisions.
(b) Volume and maximum temperature of the required cooling air and the maximum amount of dissipated heat to the ambient.
(c) Schematic diagrams with symbols, and notation to show the required voltage, current, power and tolerance of the input ship's service d.c. power.
(d) Type and size of the incoming power cable.
(e) Maximum impedance allowed for the remote voltage sensing leads.

3.9.1.3 Installation drawings. Installation drawings shall be in accordance with the data ordering document (see 6.2.2). Installation drawings shall contain the following data:

(a) Manufacturer's name and catalog number or equivalent identification of the equipment.
(b) List of descriptive data of the equipment including:
  (1) Applicable specification and exceptions.
  (2) Complete rating showing:
    a. Input voltage.
    b. Input kW, and current, at zero, one-half and rated load.
    c. Output voltage frequency and number of phases.
    d. Efficiency at one-half and rated load.
    e. Output adjustment range.
    f. Output kW, kVA, current and power factor.
    g. No load losses.
  (3) Duty (if intermittent, indicate duty cycle).
  (4) Shock classification.
  (5) Enclosure classification.
  (6) Ventilation.
  (7) Type of load.
  (8) Ambient temperature range.
  (9) Wave form of output voltage (harmonic content and deviation factor).
  (10) Weight of equipment.
  (11) Notation of the corrosion-resistant treatment of all hardware.
  (12) Special features.
(c) Outline drawing showing front, side, top and section views as necessary to present information as follows:
  (1) Overall mounting dimensions.
  (2) Location and size of mounting holes and ground stud.
  (3) Approximate arrangement and location of parts.
  (4) Location of center of gravity of equipment.
  (5) Identification and information plates.
  (6) Clearance required for ventilation and to open the drawers.
  (7) Cable entrance provisions.
  (8) Direction of air flow in ventilated equipment.
  (9) Air inlet and discharge openings.
  (10) Physical position of parts required for energization, control, protection, operation and monitoring as follows:
    a. Input line switches, contactors, circuit breakers, and auxiliary switches as necessary for control or test purposes.
    b. Fuses.
    c. Indicator lights and instruments.
    d. Test points.
(d) Location and details of terminal boards required for the connection of input line cables, output load cables, interconnection between inverters for load transfer, remote control and remote voltage sensing. The types of cables required and the maximum length allowed for interconnection between cabinets, between the equipment and the remote voltage sensing point, and between the equipment and other remote control points.
(e) Interface information showing:
  (1) Shipboard monitoring surface details and preparation, size of bolts, and tightening torques.

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(2) Ground cable and lug size and maximum tightening torque allowed.
(3) Unpacking procedure and inspection required prior to mounting.
(4) Procedure of assembly of modules, subassemblies, drawers, etc. after mounting the units to prepare for operation.

3.9.1.4 Maintenance drawings. Maintenance drawings shall be in accordance with the data ordering document (see 6.2.2). Maintenance drawings shall contain the following data:

(a) Magnetic amplifier and saturable core reactor operating characteristic curves:
(1) Plot of saturable core device output or gate winding current versus the ampere-turns developed by selected control windings, or plot of amplifier output versus control signal input for each amplifier stage.
(2) Schematic diagram of test circuit showing instrument positions and polarity of connection into test circuit.
(3) Test conditions and procedures and the expected results with acceptable tolerances.

(b) Schematic diagrams: This portion of the category II drawings shall clearly explain by means of schematic symbols, wave form and notation the electrical functions performed by the equipment, and the techniques, involving circuits, instruments, and protective devices, by which these functions are implemented and monitored. In the preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of parts and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram. Each part shall be identified by its reference designation. Terminals used for connections external to the unit shall be identified and labeled as to purpose. A notation, giving voltage, phases, frequency, adjustment range, and rated current or power shall be placed adjacent to input and output terminals, as applicable.

(c) Connection diagram showing parts and wiring in approximately the relative physical positions which they occupy in the equipment leads of parts and hook-up wire lengths used for interconnection, parts, terminals, subassemblies, and modules shall be marked. Hook-up wire size need not be shown. Markings shall be consistent with part descriptions specified elsewhere. Notes labeling terminals for external connections shall be provided on connection diagrams the same as on the schematic diagram. Wire and cable routing within the cabinet shall be indicated in terms of point of origin and destination of each conductor and cable harness, as identified in accordance with the wire markings employed in the actual equipment.

(d) Description of operation - A brief description of operation of the circuit elements to enable approval authorities and operators to understand circuitry.

(e) Test panel drawings showing circuit schematic diagram, indicator lights, test points with associated voltage wave forms, magnitude and duration.

(f) Power thyristors (semiconductor controlled rectifiers) and diodes testing methods and procedures:
(1) Tabulation of all thyristors and diodes in the power circuits.
(2) Schematic diagram of test circuits showing instrument position and polarity connection to the test semiconductors.
(3) Test conditions and procedures, required input voltage level, expected results with acceptable tolerances.

3.9.2 Technical manuals. Technical manuals shall be prepared in accordance with the data ordering document (see 6.2.2).

3.10 Workmanship. The UPS, including all parts and accessories, shall be constructed in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, marking of parts and subassemblies, wiring, welding and brazing, plating, riveting, finishes, machine operation, screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Calibration system requirements. The contractor shall provide and maintain a calibration system for test equipment. The calibration system shall be in accordance with MIL-C-45662.

4.1.2 Accuracy of test apparatus. The accuracy of instruments and test equipment used to control or monitor the test parameters, shall conform to requirements of MIL-C-45662. Instruments and test equipment used in conducting the tests specified herein shall have an accuracy of at least one-fifth the tolerance for the parameter to be measured.

4.1.3 Inspection system. The contractor shall maintain an inspection system in accordance with MIL-Q-9858 (see 6.2.1).

4.2 Test conditions. Unless otherwise specified herein, tests shall be conducted under ambient temperature 25°C ± 2°C. The specific ambient temperature at the time of test shall be measured and recorded. Transient voltage or current traces taken by oscillograph or pictures of oscilloscope shall have peak-to-peak magnitude of not less than 3 inches and the speed of the traces shall be not more than 2 cycles per inch.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.4 First article inspection. When specified (see 6.2.1), one unit of each power rating shall be subjected to the examination and tests specified in table VI.

TABLE VI. First article inspection.

Examination and test	Requirement paragraph	Inspection paragraph
General examination	3.4.1 and 3.10	4.6.1 and 4.6.2
Size and weight	3.4.3.1	4.7.1
Creepage and clearance	3.4.1	4.7.2
Insulation resistance	3.4.2.1	4.7.3
Dielectric withstanding voltage (see note)	3.4.2.1	4.7.4
Enclosures	3.3.13.3 and 3.4.3.3	4.7.5
Self-protection from input voltage variation and transient	3.3.1.1	4.7.6
Self-protection from input voltage spikes	3.3.1.2	4.7.7
Input current harmonics	3.3.1.3	4.7.8
Input current balance	3.3.1.4	4.7.9
Starting current and starting time	3.3.1.5, 3.3.1.5.1, and 3.3.1.6	4.7.10
No-load input current	3.3.1.7	4.7.11
Grounding	3.3.1.8	4.7.12
Output voltage and frequency regulation	3.3.2.1 and 3.3.2.6	4.7.13
Non-linear load	3.3.2.2	4.7.14
Output voltage remote sensing	3.3.2.3	4.7.15
Output voltage adjustment	3.3.2.4	4.7.16
Transient voltage and frequency deviation and recovery	3.3.2.5 and 3.3.2.6	4.7.17
Amplitude and frequency modulation	3.3.2.7	4.7.18
Output voltage waveform	3.3.2.8	4.7.19
Output phase displacement	3.3.2.9	4.7.20
Output voltage balance under balanced load	3.3.2.10	4.7.21
Output voltage balance under unbalanced load	3.3.2.11	4.7.22
Input voltage change effect	3.3.2.12	4.7.23
Power rating	3.3.3 and 3.3.3.1	4.7.24
Efficiency and no-load losses	3.3.4 and 3.3.5	4.7.25
Motor starting	3.3.3.2	4.7.26
Overload	3.3.6	4.7.27
Short circuit and fault current	3.3.7.2 and 3.3.7.3	4.7.28
Parallel operation	3.3.8, 3.3.8.1, and 3.3.8.2	4.7.29

See note at end of table.

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current.

TABLE VI. First article inspection. - Continued

Examination and tests	Requirement paragraph	Inspection paragraph
Automatic transfer	3.3.9	4.7.30
By-pass switch operation	3.3.10 and 3.4.10	4.7.31
Rectifier-charger and battery duty cycle	3.3.12.7	4.7.32
Leakage current	3.3.15	4.7.33
Shock	3.6.1	4.7.34
Vibration	3.6.2	4.7.35
Structureborne noise	3.6.3	4.7.36
Airborne noise	3.6.3	4.7.37
Electromagnetic interference emission and susceptibility	3.6.4	4.7.38
Temperature rise and high temperature operation	3.6.5 and 3.6.6	4.7.39
Low temperature operation	3.6.5 and 3.6.6	4.7.40
Inclined operation	3.6.8	4.7.41
Humidity	3.6.11	4.7.42
Reliability (see note)	3.7.2	4.7.43
Maintainability	3.7.3	4.7.44
Stress level conformation	3.7.4.1.3	4.7.45

Note: Any of the table VI tests may be conducted concurrently to the maximum extent possible, except that dielectric withstanding voltage test shall be conducted after all other tests but before the reliability test has been completed. The reliability test shall be conducted only after satisfactory completion of all other tests.

4.4.1 First article inspection report. A first article inspection report shall be submitted to NAVSEC for acceptance (see 6.2.2).

4.5 Quality conformance inspection. Each UPS offered for delivery shall be subjected to the quality conformance inspection specified in table VII.

TABLE VII. Quality conformance inspection.

Examination and tests	Requirement paragraph	Inspection paragraph
General examination	3.4.1 and 3.10	4.6.1 and 4.6.2
Insulation resistance	3.4.2.1	4.7.3
Self-protection from input voltage variation and transient	3.3.1.1	4.7.6
Input current harmonics	3.3.1.3	4.7.8
Input current balance	3.3.1.4	4.7.9
Output voltage and frequency regulation	3.3.2.1 and 3.3.2.6	4.7.13
Output voltage adjustment	3.3.2.4	4.7.16
Transient voltage and frequency deviation and recovery	3.3.2.5 and 3.3.2.6	4.7.17
Amplitude and frequency modulation	3.3.2.7	4.7.18
Output voltage waveform	3.3.2.8	4.7.19
Output voltage balance under balanced load	3.3.2.10	4.7.21
Output voltage balance under unbalanced load	3.3.2.11	4.7.22
Input voltage change effect	3.3.2.12	4.7.23
Overload	3.3.6	4.7.27
Short circuit and fault current	3.3.7.2 and 3.3.7.3	4.7.28
Automatic transfer	3.3.9	4.7.30
By-pass switch operation	3.3.10 and 3.4.10	4.7.31
Rectifier-charger and battery duty cycle	3.3.12.7	4.7.32
Endurance run	3.3.3 and 3.3.3.1	4.7.46
Structureborne noise	3.6.3	4.7.36
Airborne noise	3.6.3	4.7.37

4.6 General examination. Parts on assemblies and completed UPS shall be examined as specified in 4.6.1 and 4.6.2.

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4.6.1 Major parts. Examination shall be conducted to determine that:

(a) The following items conform to the applicable specification and that the size, type, and rating are as shown on the accepted drawing:
  (1) Resistors and rheostats.
  (2) Instruments, instrument transformers, shunts, and other associated accessories.
  (3) Capacitors.
  (4) Switches.
  (5) Semiconductor devices.
  (6) Terminal boards.
  (7) Laminated materials, such as glass mica glass (GMG) and glass silicon glass (GSG), used for terminal boards, sub-panels, etc.

(b) Materials used in the construction of wound components comply with the material requirements as shown on the accepted drawing.
  (1) Windings were given varnish impregnation in accordance with the procedures shown in the accepted drawing.
  (2) The resistance of windings is as shown on the accepted drawing. Number of turns shall be as specified on accepted drawing.

4.6.2 Completed equipment. Examination of the completed UPS shall be conducted to determine that:

(a) Parts, modules, and drawers are mounted as shown on the accepted drawing.
(b) Parts and modules which require servicing, repair, replacement, or periodic adjustment during the life of the equipment are readily accessible.
(c) Parts, modules, and drawers are marked with identifying symbols, and these symbols agree with those used on the accepted drawing.
(d) Terminals of the parts and modules and terminal boards are marked with identifying letters, numbers, or both, and this identification agrees with those used on the accepted drawing.
(e) Wire used for interconnecting parts, modules, and drawers are of the type shown on the accepted drawing.
(f) Wiring is neatly formed into groups and laced with flame-resistant tie or glass cord and supported or clamped in a manner which will prevent chafing of the insulation due to vibration and shock.
(g) Cable clamps used are of flame-resistant material.
(h) There are no splices in any of the wires.
(i) Wire groups from hinged doors and panels are formed and clamped in a manner that sharp bends do not occur with the panel or door in the open or closed position.
(j) Sufficient slack in wiring is allowed so that the weight of the harness is not supported by the terminal connections and so that at least two replacements can be made if lugs are clipped off at the wire end.
(k) Cable entrance provisions and connections are as shown on the accepted drawing.
(l) Sufficient space is provided for these connections and sufficient cable support within the drawers and cabinets, and connections made at the appropriate terminals.
(m) Wherever wires run through holes in metal partitions or chassis, grommets are provided for mechanical protection.
(n) Wires are not bent around sharp corners which may injure the insulation.
(o) Wires are connected by either bolted or soldered connections.
(p) Bolted connections are provided with locking devices.
(q) Both ends of all wires are marked with designations as shown on the accepted drawing.
(r) The equipment pull-out drawers have stops to lock in pull-out position.
(s) Non-identical drawers cannot be interchanged in the equipment cabinet.
(t) Identification and instruction plates are furnished as shown on the accepted drawing.

4.7 Tests.

4.7.1 Size and weight. The UPS shall be measured and weighed, and dimensions and weight recorded.

4.7.2 Creepage and clearance. Creepage and clearance distance shall be demonstrated by actual measurement.

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4.7.3 Insulation resistance tests. Insulation resistance tests shall be conducted in accordance with method 302 of MIL-STD-202 and with the battery circuit breaker open. Test conditions shall be as follows:

(a) Test potential: Test condition B (high internal impedance).
(b) Points of measurement: Between each electrically isolated circuit (see 4.7.3.1 and figure 9) and all other circuits and ground.
(c) Electrification time: 60 seconds minimum.

4.7.3.1 Electrically isolated circuits shall be determined by application of the following criteria:

(a) Circuits whose only connection to each other is by electromagnetic coupling through a magnetic core which is shared in common by the circuits shall be considered to be electrically isolated from each other.
(b) Circuits whose only connection to each other is through a capacitor shall not be considered to be electrically isolated from each other. When the purpose is to test circuits internal to the equipment, and only then, such circuits shall be temporarily interconnected with a jumper wire, or test lead.

4.7.4 Dielectric withstanding voltage tests. Dielectric withstanding voltage tests shall be conducted in accordance with method 301 of MIL-STD-202, and with the battery circuit breaker open. Test conditions shall be as follows:

(a) Magnitude of test voltage.
    (1) For circuits rated 60 volts or less, the test voltage shall be 900 volts.
    (2) For circuits rated more than 60 volts but not greater than 600 volts, the rms test voltage shall be twice rated circuit voltage plus 1000 volts.
    (3) For circuits rated above 600 volts, the rms test voltage shall be two and one-quarter times rated circuit voltage plus 2000 volts.
    (4) For circuits containing parts that are applied within their specified ratings and are in accordance with part specifications that specify a lower dielectric test voltage than specified in (1), (2), of (3), the dielectric test voltage for the circuit shall correspond to the specified voltage for the part having the lowest specified dielectric test voltage but in no case less than:
        a. 900 volts for circuits connected to the power supply terminals.
        b. 500 volts for circuits electrically isolated from the power supply terminals.
(b) Nature of test voltage. The test voltage shall approximate a true sine wave of a frequency not less than the rated frequency of the circuit under test.
(c) Points of application. The test voltage shall be successively applied between each electrically isolated circuit (see 4.7.3.1 and figure 9) and all other circuits and ground.
(d) Duration of test. The duration of the test shall be 60 seconds minimum. The test voltage shall be removed by adjustment of its value to zero, not by sudden interruption. An alternate test of 1 second minimum using a touch probe may be substituted if a test voltage 20 percent higher than that specified in (a) is used, and a determination has been made that no surges which could damage parts are produced by sudden interruption of the test voltage.
(e) Definition of failure. Any evidence of arcing, flashover, or punctured insulation shall be interpreted as a failure of the test. Corona (visible, audible, or odorous) shall not be considered as a failure.

4.7.5 Enclosures. Acceptability of the enclosures shall be determined by performance of the dripproof test of MIL-STD-108.

4.7.6 Self-protection from input voltage variation and transient. The test shall be performed as follows:

(a) The UPS shall be operated with rated input voltage at no-load and rated-load, 0.8 p.f., respectively. The input voltage of the UPS shall be raised and then lowered beyond the specified operation range for 3 seconds and 50 ms, respectively. The input voltage, output voltage, and output current shall be recorded with an oscillograph.

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(b) The UPS shall be operated at rated input voltage to supply rated-load. The input power shall be interrupted for a period of 30 seconds and then restored to the original condition. The test shall be repeated by interrupting the input power for periods of 100 and 50 ms. The UPS input voltage, output voltage, and output current shall be recorded with an oscillograph.

4.7.7 Self-protection from input voltage spikes. The input to the UPS shall be rated voltage having a source impedance of 0.4 ohm inductive reactance (at 60 Hz) and 0.07 ohm resistance. The UPS shall be disconnected from the input lines, and applicable transient voltage as specified in 3.3.1.2 shall be applied between line-to-line and between line-to-ground. After reconnecting the UPS back to the input lines, the transient voltage shall be reapplied as previously established when the UPS is operated at no-load and rated-load, 0.8 p.f. The UPS output voltage shall be observed on an oscilloscope. Pictures of the voltage waveforms shall be taken.

4.7.8 Input current harmonics. The UPS shall be operated at rated input voltage and frequency. The input current harmonics of each input line shall be determined at no-load, 50 percent load, and rated-load. Harmonics less than 0.1 percent may be disregarded. The current harmonics shall be measured by means of shunts connected in each input line to the UPS. The fundamental and harmonics of the voltage drops across the shunts shall be determined by a harmonic analyzer.

4.7.9 Input current balance. Rated input voltage to the UPS shall be applied with 3 percent unbalance among the three phases. The UPS shall be operated at 10 percent load, 50 percent load and rated-load at rated output voltage and frequency. The input current of each line to the UPS shall be measured with indicating meter having initial accuracy of 1 percent or better.

4.7.10 Starting current and starting time. The rated voltage shall be connected to the UPS input, and the UPS shall be started with no-load at the output. The input starting current and output voltage shall be measured with an oscillograph. The test shall be repeated with full-load connected to the UPS. The UPS start-up shall be attempted by setting the input voltage 2 volts beyond the specified limits as specified in 3.3.1.5.1. The starting time and starting current shall be determined.

4.7.11 No-load input current. The UPS shall be operated at no-load with the battery fully charged. The input current shall be measured for the maximum, rated, and minimum input voltages.

4.7.12 Grounding. The test shall be performed with the UPS operated at rated input voltage and frequency at no-load and at rated-load. Each input and output terminal shall be grounded in turn to ground as follows and pictures of the input and output voltage waveforms shall be taken by oscilloscope or oscillograph:

(a) Connect each terminal to ground through a 100 kilohm resistance.
(b) Momentarily ground each terminal in turn by shorting one of the 100 kilohm resistance in (a). Remove that short and then short the other 100 kilohm resistance.

4.7.13 Voltage and frequency regulation. The UPS shall be connected to an input voltage of which can be varied as shown in table VIII. After the UPS has been stabilized at rated-load, the output steady-state voltage and frequency regulation shall be determined for all conditions specified in table VIII.

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TABLE VIII. Conditions of input and output.

Test condition numbers	Input						Output (at rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Percent of rated-load				P.f. of load			
	Maximum	Rated	Minimum	Maximum	Rated	Minimum	0 percent	50 percent	100 percent	125 percent	0.6 lagging	0.8 lagging	1.0	0.9 leading
1	X	-	-	X	-	-	X	-	-	-	-	-	-	-
2	-	X	-	-	X	-	X	-	-	-	-	-	-	-
3	-	-	X	-	-	X	X	-	-	-	-	-	-	-
4	X	-	-	X	-	-	-	X	-	-	X	-	-	-
5	-	X	-	-	X	-	-	X	-	-	X	-	-	-
6	-	-	X	-	-	X	-	X	-	-	X	-	-	-
7	X	-	-	X	-	-	-	X	-	-	-	X	-	-
8	X	-	-	-	X	-	-	X	-	-	-	X	-	-
9	-	X	-	-	X	-	-	X	-	-	-	X	-	-
10	-	-	X	-	X	-	-	X	-	-	-	X	-	-
11	-	-	X	-	-	X	-	X	-	-	-	X	-	-
12	X	-	-	X	-	-	-	X	-	-	-	-	X	-
13	X	-	-	-	X	-	-	X	-	-	-	-	X	-
14	-	X	-	-	X	-	-	X	-	-	-	-	X	-
15	-	-	X	-	X	-	-	X	-	-	-	-	X	-
16	-	-	X	-	-	X	-	X	-	-	-	-	X	-
17	X	-	-	X	-	-	-	X	-	-	-	-	-	X
18	-	X	-	-	X	-	-	X	-	-	-	-	-	X
19	-	-	X	-	-	X	-	X	-	-	-	-	-	X
20	X	-	-	X	-	-	-	-	X	-	X	-	-	-

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TABLE VIII. Conditions of input and output. - Continued

Test condition numbers	Input						Output (at rated voltage and frequency)							
	Voltage within the specified range			Frequency within the specified range			Percent of rated-load				P.f. of load			
	Maximum	Rated	Minimum	Maximum	Rated	Minimum	0 percent	50 percent	100 percent	125 percent	0.6 lagging	0.8 lagging	1.0	0.9 leading
21	-	X	-	-	X	-	-	-	X	-	X	-	-	-
22	-	-	X	-	-	X	-	-	X	-	X	-	-	-
23	X	-	-	X	-	-	-	-	X	-	-	X	-	-
24	X	-	-	-	X	-	-	-	X	-	-	X	-	-
25	-	X	-	-	X	-	-	-	X	-	-	X	-	-
26	-	-	X	-	X	-	-	-	X	-	-	X	-	-
27	-	-	X	-	-	X	-	-	X	-	-	X	-	-
28	X	-	-	X	-	-	-	-	X	-	-	-	X	-
29	X	-	-	-	X	-	-	-	X	-	-	-	X	-
30	-	X	-	-	X	-	-	-	X	-	-	-	X	-
31	-	-	X	-	X	-	-	-	X	-	-	-	X	-
32	-	-	X	-	-	X	-	-	X	-	-	-	X	-
33	X	-	-	X	-	-	-	-	X	-	-	-	-	X
34	-	X	-	-	X	-	-	-	X	-	-	-	-	X
35	-	-	X	-	-	X	-	-	X	-	-	-	-	X
36	-	X	-	-	X	-	-	-	-	X	-	X	-	-
37	-	X	-	-	X	-	-	-	-	X	-	-	X	-

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4.7.14 Non-linear load. The UPS shall be operated with the rated input voltage and frequency. The output of the UPS shall be connected to resistive load through a full-wave rectifier bridge. The test shall be conducted with rated-load and with 50 percent rated-load. The UPS output waveform, voltage regulation, and amplitude modulation shall be measured and recorded.

4.7.15 Output voltage remote sensing. The UPS shall be connected to rated-load with conductors which have a voltage drop of 7 percent of the rated output voltage at 200 percent of rated current. With the UPS operated at rated-load, 0.8 p.f., the voltage at the remote sensing point shall be adjusted to the rated voltage by means of the voltage adjustment. Further adjustment will not be allowed for the rest of this test. The voltage regulation, amplitude modulation, and voltage waveform at the UPS output terminal and at the remote sensing point shall be measured for test conditions 2, 5, 9, 14, 18, 21, 25, 30, 34, and 36 of table VIII. The loss of remote sensing signal shall be simulated for each test condition. The voltage adjustment test shall be repeated as specified in 4.7.16 to demonstrate the required voltage range at the remote sensing point. For the 3-phase UPS, the unbalanced load test as specified in 4.7.22 shall also be repeated, except that unbalanced loads shall be applied at the remote sensing point.

4.7.16 Output voltage adjustment. After the UPS has been stabilized at rated-load, the output voltage at the UPS output terminals shall be adjusted over the full range using the voltage adjustment device. The test shall be conducted for test conditions 2, 5, 9, 14, 18, 21, 25, 30, 34, and 36 of table VIII.

4.7.17 Transient voltage, frequency deviation, and recovery. The UPS shall be operated with rated input. Change balanced loads of 0.8 and unity p.f. in one step as follows:

(a) No-load to 50 percent.
(b) 50 percent to rated-load.
(c) Rated-load to 50 percent.
(d) 75 percent to 25 percent.
(e) No-load to rated-load.
(f) Rated-load to no-load.

Oscillograms or oscilloscope pictures shall be taken on the load application and load removal to determine the voltage transient and recovery time. This test shall be conducted for input conditions 8, 9, and 10 as specified in table VIII.

4.7.18 Amplitude and frequency modulation. The UPS output voltage amplitude and frequency modulation shall be determined for test condtions 2, 5, 8, 9, 10, 13, 14, 15, 18, 21, 24, 25, 26, 29, 30, 31, and 36 of table VIII.

4.7.19 Output voltage waveform. The harmonic content, deviation factor, and voltage spikes of the UPS output voltage shall be determined for test conditions 2, 5, 8, 9, 10, 13, 14, 15, 18, 21, 24, 25, 26, 29, 30, 31, and 36 of table VIII.

4.7.20 Output phase displacement angle. For the 3-phase UPS, the output displacement angle shall be measured at the output terminals for test conditions 2, 9, 14, 18, 25, and 30 of table VIII.

4.7.21 Output voltage balance under balanced load. For the 3-phase UPS, the output voltage of each phase at the output terminals shall be recorded for test conditions 2, 5, 8, 9, 10, 13, 14, 15, 18, 21, 24, 25, 26, 29, 30, 31, and 36 of table VIII.

4.7.22 Output voltage balance under unbalanced load. For the 3-phase UPS, the UPS shall be operated with rated input. A 3-phase, 0.6 p.f. load shall be applied across the output phases to have the lines carry current of 75 percent of rated current. A single-phase 0.6 p.f. load shall be applied across the UPS output phase AB to have line A carry an additional current of 15 percent of rated current. The output voltage of all phases shall be recorded at the output terminals. The test shall also be conducted using a single-phase 0.6 p.f. load across output phase BC and CA. The UPS shall be operated at rated input. A single-phase, 0.6 p.f. load shall be applied across output phases AB to have line A carry a current of 15 percent of the rated current. The output voltages of all phases shall be recorded. The test shall also be conducted using a single-phase 0.6 p.f. load across output phases BC and then CA. The above tests shall be repeated for 0.8 p.f. load.

4.7.23 Input voltage change effect. The input to the UPS shall be set at test conditions 1, 25, and 27 of table VIII, and the input voltage shall instantaneously be increased to the limits and duration as specified in 3.3.2.12, and then returned to the initial value. The input and output voltage transient and recovery time shall be recorded and determined

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by an oscillograph. These tests shall be repeated with the input voltage instantaneously decreased to the limits and duration as specified in 3.3.2.12, and then returned to the initial value.

4.7.24 Power rating. The UPS shall be operated at rated input voltage and frequency for 2 hours each of the following loads:

(a) Rated-load, 0.6 p.f. lagging.
(b) Rated-load, 0.8 p.f. lagging.
(c) Rated-load unity p.f.
(d) Rated-load 0.9 p.f. leading.

The output voltage shall be recorded at the beginning and at the end of each load operation and the voltage regulation shall remain within the limits specified in 3.3.2.1.

4.7.25 Efficiency and no-load losses. The battery of the UPS shall be disconnected. Input and output voltages and frequency of the UPS shall be set at rated values. Input and output power shall be measured for the following load conditions:

(a) Rated-load, 0.8 p.f. lagging.
(b) Rated-load, unity p.f.
(c) 50 percent load, 0.8 p.f. lagging.
(d) 50 percent load, unity p.f.
(e) No-load.

The UPS no-load losses and efficiencies at 50 percent load and rated-load shall be determined.

4.7.26 Motor starting. The UPS shall be operated at a rated input voltage. A 0.3 lagging p.f. static load shall be connected to the UPS through a circuit breaker or contactor to simulate a motor starting condition. The static load shall draw 200 percent of the UPS rated current. The circuit breakers or contactors shall be closed for a period of 30 seconds, and the UPS output voltage and current shall be recorded with an oscillograph.

4.7.27 Overload. The UPS shall be operated at rated input voltage and frequency. The UPS shall be overloaded to 200 percent of the rated-load at rated output voltage, 0.8 p.f., for more than one minute. The UPS output voltage, output current, and time required to transfer the UPS shall be recorded with an oscillograph. The load shall be reduced from 200 percent to 50 percent of rated-load to test the UPS retransfer operation.

4.7.28 Short circuit and fault current. A 3-phase short circuit fault (for the 3-phase UPS) and single-phase short circuit fault (for both 3-phase and single-phase UPS) shall be applied to the UPS output. This test shall be conducted with the maximum, rated, and minimum input voltages in the specified range. The fault current and the time required to transfer the UPS load shall be recorded with an oscillograph. After the load has been transferred to the UPS input a.c. lines, the short circuit shall be removed to test the UPS retransfer operation. This test shall be repeated with the fault current of 210 percent of the rated current.

4.7.29 Parallel operation. After two units of the UPS have been automatically synchronized to operate in parallel, no further adjustment shall be made for the remainder of the test. The total load of the paralleled units shall be varied from 10 to 90 percent of the combined kW and kVA rating of the two units at 0.8 p.f. in 25 percent load steps and back to 10 percent load in the same manner. The units shall operate for at least 15 minutes at each load step. The difference in kW and kVA supplied by each unit shall be recorded and computed. For each load step, the parallel test shall be conducted at the maximum, rated, and minimum input voltage in the specified range. Simulated internal fault shall be applied to one of the paralleled units to demonstrate selective tripping. The UPS output voltage shall be recorded by an oscillograph or picture taken from an oscilloscope.

4.7.30 Automatic transfer. The UPS shall be operated with rated input voltage, frequency, and rated-load. The automatic transfer switch operation, transfer time, and voltage synchronization shall be determined with an oscillograph for the following malfunctions:

(a) UPS output voltage outside the regulation range as specified in 3.3.2.1.
(b) UPS output frequency outside the regulation range as specified in 3.3.2.6.
(c) The loss of d.c. input to the inverter.

Each malfunction shall be corrected to determine the retransfer of loads. The test shall be repeated for 50 percent of rated-load condition.

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4.7.31 By-pass switch operation. The by-pass switch manual operation test shall be conducted with the UPS at rated input voltage and the output loads at no-load, 50 percent load, and rated-load. When the UPS is operated with rated input voltage and at no-load, the input frequency shall be increased and decreased to beyond the plus or minus 5 percent type I regulation limits for a period of several seconds and then return to the rated value. Input and output frequencies of the UPS shall be recorded by a dual-channel memory oscilloscope, or equivalent instrument, to determine the frequency synchronization.

4.7.32 Rectifier-charger and battery duty cycle. The UPS output power, current, voltage, and harmonics shall be recorded as indicated below, and voltage regulation shall be calculated when the UPS is operated with the following duty cycle:

(a) Set the UPS input voltage and frequency at rated value and the output at rated-load, 0.8 p.f. The battery shall be fully charged and the specific gravity of the electrolyte shall be recorded. Record instrumentation readings.
(b) Interrupt the UPS input power for a period of 15 minutes. Record instrumentation readings at the beginning and at the end of this 15 minute period.
(c) Reapply the UPS input power and maintain at step (a) value for 1 hour. Record instrumentation readings at the beginning and at the end of this 1-hour period.
(d) Repeat steps (b) and (c).
(e) Run the UPS for a period of 8 hours with the input power maintained as step (a). Record instrumentation readings every hour.

Upon completion of the above duty cycle, the specific gravity of the battery shall be recorded and the UPS output voltage transient and recovery test shall be conducted as specified in 4.7.17 for input condition 9, as specified in table VIII, only.

4.7.33 Leakage current. UPS input and output EMI capacitors (if they are connected to the UPS ground stud) shall be disconnected from the ground stud. A low resistance shunt shall be connected between the EMI capacitor, one phase at a time, and the ground. The UPS shall be operated with the rated input voltage. The ground current through the shunt shall be measured with a waveform analyzer.

4.7.34 Shock. High-impact (HI) shock tests (electrical operation and mechanical damage) shall be conducted in accordance with MIL-S-901 and as specified in 4.7.34.1:

4.7.34.1 Test features. Test features shall be as follows:

(a) Type A (for principal units) test shall be performed. The UPS shall be energized and operated at rated-load during shock test.
(b) Definition of failure to perform principal functions shall be as follows:
    (1) Breakage of any parts, including mounting bolts. Minor chipping of parts such as plastic knobs and cases, and minor distortion of parts will be permitted where such chipping or distortion cannot in any manner impair operation of the UPS as specified.
    (2) Appreciable distortion of any parts, including enclosure and framework.
    (3) A value of insulation resistance lower than that permitted by this specification (see 3.4.2.1).
    (4) Low dielectric strength. After shock tests the dielectric test shall be conducted at a voltage equal to 65 percent of the voltage specified in 4.7.4. Failure to pass this test shall be cause for rejection.
    (5) Failure to pass visual examination. The UPS shall be carefully examined after removing removable panels, doors, and drawers to ascertain any mechanical damage. If necessary, partial disassembly shall be performed to aid in determining any possible damage.
    (6) Failure to continually carry rated-load during shock test and to perform electrical test following shock. Adjustment or replacement of damaged parts shall not be permitted during shock test.
(c) Mounting. The UPS shall be mounted on the shock machines in a manner simulating shipboard installations in accordance with MIL-S-901 for deck-mounted equipment.
(d) Immediately following the shock test, the UPS shall be tested for output voltage and frequency regulation (see 4.7.13), and transient voltage, frequency deviation, and recovery (see 4.7.17) of rated voltage and frequency.

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4.7.35 Vibration. Vibration measurements shall be obtained in accordance with the measurement and data reporting procedures as specified in MIL-STD-167-1.

4.7.35.1 Test features. Test features shall be as follows:

(a) The UPS shall be mounted so as to simulate actual shipboard installation.
(b) The UPS shall be energized and operated at rated-load during vibration tests.
(c) The failure determination as specified for the shock tests in 4.7.34.1(b) shall apply also to the vibration tests.
(d) Immediately following the vibration test, the UPS shall be tested for output voltage and frequency regulation (see 4.7.13), and transient voltage, frequency deviation, and recovery (see 4.7.17) with rated input voltage and frequency.

4.7.36 Structureborne noise. Structureborne noise measurements shall be obtained in accordance with the measurement and data reporting procedures as specified in MIL-STD-740. UPS units shall be operated at rated-load during the noise tests. Reports on the noise test shall be provided in accordance with the data ordering document (see 6.2.2).

4.7.37 Airborne noise. Airborne noise measurements shall be obtained in accordance with the measurement and data reporting procedures as specified in MIL-STD-740, except that noise levels shall be as specified in table V (see 3.6.3). UPS units shall be operated at rated-load during the noise tests. Reports on the noise test shall be provided in accordance with the data ordering document (see 6.2.2).

4.7.38 Electromagnetic interference emission and susceptibility. An electromagnetic interference emission and susceptibility test shall be conducted in accordance with MIL-STD-462. Tests shall be conducted on each input and output conductor and remote control circuit conductor of the UPS. Data to be recorded shall be in accordance with MIL-STD-461.

4.7.39 Temperature rise and high temperature operation. Temperature rise tests shall be conducted at an ambient temperature of 50°C. The test methods to be employed and the precautions to be observed shall be as specified in 4.7.39.1 through 4.7.39.5.

4.7.39.1 Assembly and mounting of equipment. Temperature rise tests shall be made only with the UPS completely assembled and mounted in the manner for which it is designed. Barriers shall be placed adjacent to the enclosure, at the minimum clearance specified on the contractor's drawing for adequate ventilation, to simulate shipboard space restrictions. Temperature rise measurements shall be made on all points as specified on contractor prepared drawings which shall be submitted with a test plan (see 6.2.2). The drawings shall show the locations, types, and methods of installation of all temperature sensors. The minimum temperature measurement points shall be as follows:

(a) Power semiconductor studs.
(b) Capacitors.
(c) Power transformer and reactor coil windings.
(d) Ambient temperature.
(e) Cooling air inlet and outlet.
(f) Battery electrolyte.

4.7.39.2 Method of temperature measurements. The methods of temperature measurement of ambient and parts or subassemblies shall be in accordance with MIL-E-917, except that thermometers shall not be used.

4.7.39.3 Method of loading. The UPS shall be operated at test conditions 25 and 30 of table VIII. The temperatures shall reach the steady final value as specified in 4.7.39.4 for each input voltage and for each load condition. Temperature rises shall be in accordance with temperature requirements of MIL-E-917.

4.7.39.4 Duration of test. The consecutive temperature measurement of each part or module shall be recorded not more than 5 minutes apart. The temperature rise test shall be continued until the temperature rise of parts or modules recorded during test have attained a steady final value. Steady final value is that value of temperature which will remain within a 2°C band for a period of 2 hours while the ambient temperature is maintained at 50°C.

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audible, or odorous, shall not be considered as a failure.

4.7.39.5 High temperature operation. After the UPS has attained a steady final value as specified in 4.7.39.4, the following tests shall be conducted at the ambient temperature 50°C, except that load conditions shall be as specified in 4.7.39.3:

(a) Voltage and frequency regulation test as specified in 4.7.13.
(b) Voltage adjustment test as specified in 4.7.16.
(c) Transient voltage, frequency deviation, and recovery test as specified in 4.7.17.
(d) Amplitude and frequency modulation test as specified in 4.7.18.
(e) Output voltage waveform test as specified in 4.7.19.
(f) Short circuit protection test as specified in 4.7.28.
(g) Input voltage step change test as specified in 4.7.23.

4.7.40 Low temperature operation. Prior to this test, the UPS shall be sufficiently conditioned to assure that each part is at a temperature of 0°C to 5°C. Rated input shall be connected to the UPS. The UPS shall be started at an ambient temperature of 0°C and shall operate at no-load and rated-load 0.8 p.f. for a period of 1 hour at each load condition.

4.7.41 Inclined operation. The UPS shall be tested for inclined operation at 45 degrees from the horizontal. The UPS shall be run at rated-load for a period of not less that 1 hour at each of the following positions:

(a) 45 degrees forward.
(b) 45 degrees backward.
(c) 45 degrees to left side.
(d) 45 degrees to the right side.

During the progress of this test, it shall be ascertained that there is no apparent overheating of any drawers, modules, or parts; dropping out of contactors; or malfunction of operation of the UPS.

4.7.42 Humidity. The UPS shall be tested as specified in 4.7.42.1 and 4.7.42.2.

4.7.42.1 Procedure. Transformer and reactor windings shall be disconnected as necessary from other components to permit accurate measurements of insulation resistance dissipation factor of the winding and ground insulation. Initial measurements of insulation resistance and dissipation factor shall be taken at the temperature and relative humidity of the surrounding air. The wound component shall next be conditioned to arrive at a dry condition. Conditioning shall be obtained by placing the wound component in a heat chamber for drying over a period of 6 hours with temperature maintained at 105°C. At the end of this period, the wound component shall be removed, and after cooling (within 8 hours) the insulation resistance and dissipation factor shall again be measured. The UPS shall then be immediately placed in the humidity chamber for a period of one week (168 hours). While under humidification, daily measurements of ambient temperature, insulation resistance, and dissipation factor shall be made and shall meet the values specified in figures 10 and 11. At the end of the seventh day of the humidification period, and within 5 minutes after removal from the humidity chamber, the insulation resistance and dissipation factor shall be measured. Following measurements, the equipment shall be carefully examined for any signs of material damage and corrosion. The extent of any observed damage or corrosion shall be noted and cause determined if feasible. After examination, any disconnections or disassembly shall be reverted to original condition, and the UPS output voltage and frequency regulation and voltage adjustment test as specified in 4.7.13 and 4.7.16 shall be rerun. Following operation, insulation resistance shall be measured and recorded between isolated circuits and between circuits and ground.

4.7.42.2 Recommended test apparatus. Recommended test apparatus shall be as follows:

(a) The humidity chamber can be made of steel or by placing a transparent plastic film having a low moisture permeability over a steel framework. The top of the chamber shall be slanting or peaked (or otherwise arranged) so that the excess condensate does not drip on the equipment. The humidity shall be maintained at 100 percent relative humidity by electrically heating an open pan of water within the chamber. The amount of exposed surface of the water shall be sufficient to produce minute droplets of condensate on the equipment and insulation surfaces under test. However, the amount of condensate shall be controlled so as not to produce puddles or streams of water on the insulation surfaces. The drawers and panels shall be opened to permit free access of the humidified air to all parts and surfaces.

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(b) The insulation resistance measuring equipment shall be similar and equivalent in accuracy to the General Radio Megohm Bridge, type 544-B, 500 volt d.c., with a range from 0.1 to 1,000,000 megohms.
(c) The dissipation factor measuring equipment shall be similar and equivalent in accuracy to the General Radio Capacitance Test Bridge, type 1611-A, 60 Hz, with a range of dissipation factor from 0 to 50 percent.

4.7.42.3 Methods of measurement. Methods of measurement shall be as follows:

(a) The insulation resistance measurement shall be made using a potential of 500 volts d.c. applied for a 1 minute period. The insulation resistance shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground, and between insulated windings.
(b) The dissipation factor shall be measured with the capacitance test bridge with values read directly in percent from the bridge. Dissipation factor shall be measured on all transformers and reactors. Measurements shall be made and recorded between each winding and ground, and between insulated windings.

Note: Due to the need for exposing copper for the connections to the windings, a leakage path is produced by humidification over the various insulation surfaces between the base copper and the other parts of the equipment that are otherwise insulated from the copper circuit. Therefore, leads should be held separated as much as possible and covered with a nonwetting grease which shall be wiped off with a clean dry rag upon removal from humidification. Measurements shall be made either with UPS in the humidity chamber or within minutes after removal of the UPS from the humidity chamber.

4.7.43 Reliability test. The reliability test shall be conducted in accordance with MIL-STD-781 (see 3.7.2), except as otherwise specified herein. The test plan shall be in accordance with MIL-STD-781 having producer and consumer risks of 20 percent and discrimination ratio of 3.

4.7.43.1 Reliability test conditions. The following reliability test conditions shall apply instead of the test levels in MIL-STD-781:

(a) Ambient temperature - 45°C to 50°C.
(b) Ambient temperature cycling - None.
(c) Vibration - None.
(d) Output load cycling - During each 8-hour day shift, the UPS shall be cycled 3 times. Each load cycle shall consist of the following:
    (1) 1-1/2 hours at rated-load.
    (2) 20 minutes at no-load.
    (3) 10 minutes shut-down.
    During the remainder of each 24 hour day, except for the additional test specified in (1), the UPS shall be operated at rated load.
(e) Output voltage - Rated output voltage.
(f) Output frequency - Rated output frequency.
(g) Input voltage - For the load cycle during 8-hour day shift as specified in (d), the input voltage shall be as follows:
    (1) The first load cycle - 5 percent below the rated voltage.
    (2) The second load cycle - Rated voltage.
    (3) The third load cycle - 5 percent above the rated voltage.
    For the remainder of each 24 hour day, the input voltage shall be rated input plus or minus 5 percent.
(h) Input frequency - 60 Hz.
(i) Additional test - After the third load cycle of each 8-hour day shift, the output voltage transient shall be conducted as specified in 4.7.17(e) and (f). The output voltage regulation and modulation shall be conducted as specified in 4.7.13 and 4.7.18. For these additional tests, the rated input and output voltage and frequency shall be used, and the load conditions shall be no-load, 50 percent load, and rated-load, 0.8 p.f. lagging. The input current, input voltage, input frequency, output current, output frequency, output voltage, and output power shall be recorded at rated-load and at no-load for each load cycle during each 8-hour day shift. For the remainder of each 24-hour day, these data shall be recorded by pen and ink type, or equivalent, recorders.

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4.7.43.2 Failure. The definition of failure shall be in accordance with 3.7.2.1.

4.7.43.3 Preventive maintenance. Preventive maintenance during the reliability test shall be limited to the type and amount of preventive maintenance required by the accepted technical manual.

4.7.44 Maintainability demonstration test. The UPS shall be subjected to maintainability demonstration test as follows:

(a) For substitution of drawers (see 3.7.3.1(a)), and for substitution of circuit boards and parts within drawers (see 3.7.3.1(b)), the demonstration of M_{ct} shall be in accordance with MIL-STD-471, method 2, at consumer's risk of 20 percent (β = .84), and M_{maxct} at the 90th percentile (θ = 1.282). After completion of each maintenance action involving replacement of a drawer, the equipment shall be operated for 15 minutes at rated-load.

(b) Circuit boards and modules after removal from drawers (see 3.7.3.1(c)). Demonstration of equipment repair time shall be in accordance with MIL-STD-471, method 3.

4.7.44.1 Faults for correction. Corrective maintenance tasks for the demonstration test shall consist of the correction of simulated faults in the UPS. No advance information on the faults shall be allowed to the demonstration technicians.

4.7.44.2 Scope of demonstration tasks. The time measured for each corrective maintenance task shall consist of the sum of the times of all the steps sufficient to complete the repair, including preparation, fault location, fault correction, adjustment, calibration checkout, and the consulting of technical manuals and other allowed aids.

4.7.44.2.1 Test for replacement of drawers. Faults shall be simultated in each drawer of each cabinet, one or more drawers at a time.

4.7.44.2.2 Test for repair of drawers or modules. Each type of drawer or module, or printed circuit board shall have simulated faults. Test equipment in accordance with 3.7.3.3 shall be used to detect the simulated fault and to repair the repairable printed circuit boards and repairable modules.

4.7.45 Stress level conformation. The electrical and thermal stresses of the parts in the UPS shall be tested to conform to the calculated stresses which have been used for the calculation of the predicted MTBF (see 3.7.2). Stress level shall be tested on representative parts. The list of representative parts shall be provided with the test plan by the contractor. If any stress discrepancy of more than 10 percent is found between the tested and calculated values, the stress of that particular part and the predicted MTBF shall be corrected.

4.7.46 Endurance run. A continuous 168-hour endurance run shall be conducted on the UPS. The test condition shall be as follows:

(a) Ambient temperature - 45°C to 50°C.
(b) Output load cycling - During each 8-hour day shift, the UPS shall be cycled 3 times. Each load cycle shall consist of the following:
    (1) 1.5 hours at rated-load.
    (2) 20 minutes at no-load.
    (3) 10 minutes shut-down.
    During the remainder of each 24-hour day, the UPS shall be operated at rated-load.
(c) Output voltage - Rated voltage.
(d) Output frequency - Rated frequency.
(e) Input voltage - Rated voltage.
(f) Input frequency - Rated frequency.
(g) Data - The input current, input voltage, input frequency, output current, output frequency, output voltage, and output power shall be recorded for each load cycle during each 8-hour day shifts. For the remainder of each 24-hour day, these data shall be recorded by pen and ink type, or equivalent, recorders.
(h) Additional tests - After completion of the 168-hour run, the output voltage transient test shall be conducted as specified in 4.7.17(e) and 4.7.17(f). The output voltage regulation and modulation tests shall be conducted as specified in 4.7.13 and 4.7.18. For these additional tests, the rated input and output voltage and frequency shall be used and the load conditions shall be no-load, 50 percent load, and rated-load, 0.8 p.f. lagging.

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4.7.46.1 Failure. The definition of failure shall be in accordance with 3.7.2.1. A failure is not considered to have occurred if malfunction can be attributed to external causes such as operator's error or ambient temperature over 50°C.

4.8 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation-packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing, and marking. The UPS, technical manuals, and data shall be preserved-packaged level A or C, packed level A, B, or C as specified (see 6.2.1) and marked in accordance with MIL-E-17555. Packaging and packing drawings and test report as specified in MIL-E-17555 shall be furnished in accordance with the data ordering document (see 6.2.2).

5.1.1 Special marking. In addition to the marking requirements of 5.1, the UPS tested and certified (see 3.6.3, 4.7.36 and 4.7.37) shall have these shipping containers marked in accordance with MIL-STD-740.

5.2 Repair parts and tools. Repair parts and tools shall be preserved-packaged, packed, and marked in accordance with MIL-E-17555 or PPP-P-40 as applicable with the level specified for the intended use and destination as follows:

Destination	Level	
	Preservation-packaging	Packing
Onboard (accompanying the UPS)	A	C
Stock	A	B
Immediate use	C	C
INCO kits	A	C

Unless otherwise specified (see 6.2.1), repair parts shall be packaged one part to a unit package, except when used in quantities greater than one or parts comprising a single set or assembly. Parts, sets, or assemblies shall be individually protected to prevent direct surface contact with surfaces of adjacent parts and shall be packaged together as a single unit.

5.2.1 Repairables. Repair parts subject for return to a repair facility for restoration and reissue shall be packaged and packed in materials and containers capable of reuse. Unless otherwise specified (see 6.2.1), packaging shall be in accordance with MS90363. Repairables requiring reusable containers, such containers shall be in accordance with the guidelines of MIL-E-17555. Containers shall be marked: "REUSABLE DO NOT DESTROY". Packaging and transportation support data shall be furnished for items falling in this category (see 6.2.2).

5.2.2 Sensitive electronic item protection. Items such as, but not limited to, diodes, transistors, integrated circuits, and equipments incorporating such items which are susceptible to damage from electrostatic, electromagnetic, magnetic, or radioactive field forces shall be protected against such damage as specified in 5.2.2.1 through 5.2.2.7.

5.2.2.1 General requirements.

5.2.2.1.1 Lead and terminal protection. Leads or terminals configuration shall be maintained as manufactured without causing loads or stresses capable of causing damage to the item. Protection shall be by means of a carrier, container design, or inserts of suitable noncorrosive electrostatic-free supporting materials. Materials used to maintain item position and lead or terminal configuration shall permit item removal and replacement without damage to the item.

5.2.2.1.1.1 Carrier. Carriers, when used for additional protection of miniature electronic items, shall be of sufficient strength to withstand vibration and shocks expected in the handling and transportation environment without introducing damaging resonances, shocks, and electrostatic charges to the sensitive item(s). Anchoring or securing of the item, leads or terminals within the carrier by means of tape or adhesive is prohibited. The

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carrier shall maintain physical separation and manufactured configuration of the item leads or terminals during packaging, handling, transportation, storage/stowage and for testing operations. The carrier shall be designed to permit safe and easy removal, inspection, and item replacement, and designed without sharp edges to preclude subsequent damage to the item and packaging material/method.

5.2.2.1.2 Wraps and cushioning. Wraps and cushioning, when required for additional protection, shall be noncorrosive and in compliance with the requirements of MIL-P-116, and shall not crumble, flake, powder, or shed. Wraps or cushioning in direct contact with electrostatic sensitive item(s) shall conform to the electrostatic protection requirements specified herein.

5.2.2.2 Specific requirements.

5.2.2.2.1 Electrostatic protection. Items adversely affected by electrostatic field forces shall be wrapped in material conforming to MIL-B-81705, type I or II, or cushioned in material conforming to PPP-C-1842, type III, style A or B, and unit packaged in heat-sealed bags conforming to MIL-B-117, type I, class A, style 2 or type I, class F, style 1. Alternatively, reclosable cushioned pouches conforming to MIL-P-81997, type I or II may be used in lieu of the initial wrap or cushioning. Noncorrosive conductive material(s) shall be applied to all exposed leads and connector pins to maintain a common potential. This is to protect the item(s) from electrostatic charge(s) that may be encountered during handling. Packaging materials currently covered by title, scope, or intended use under Government specification(s), but modified as electrostatic-free material(s), or newly developed electrostatic-free packaging material(s) not covered by a Government packaging material specification(s) are encouraged for use. Use of such modified or newly developed electrostatic-free packaging material(s) will be permitted subject to the contracting officer's determination that:

(a) The physical properties of such material(s) are equal to or better than similar constructed material(s) covered under a required Government packaging material specification.
(b) Such materials satisfy the electrostatic decay rate requirement of MIL-B-81705.

The material manufacturer/contractor or contractor shall furnish to the Government inspector for review documented proof of conformance to the requirements specified herein, certified by an acceptable testing laboratory. Upon submission for acceptance, copies of the contractor's documented proof shall be forwarded to the acquisition contracting officer packaging activity. The decision of the contracting officer shall be final as to the acceptability or nonacceptability of the packaging material, and that decision shall not be subject to review under the disputes clause of the contract.

5.2.2.3 Electromagnetic protection. Unless otherwise specified in the contract or order, item(s) subject to damage by electromagnetic forces shall be unit packaged in heat-sealed, barrier bags conforming to MIL-B-117, type I or II, class E, style 1 or type I, class F, style 1. When MIL-B-117, type I or II, class E, style 1 bags are selected and used, the barrier material shall also contain a laminate of aluminum foil as well as meeting the requirements of MIL-B-117.

5.2.2.4 Electromagnetic and electrostatic protection. When the item(s) requires both electromagnetic and electrostatic protection, packaging shall be in heat-sealed, barrier bags conforming to MIL-B-117, type I, class F, style 1.

5.2.2.5 Magnetic protection. Protection against magnetic fields (as opposed to electrostatic or electromagnetic forces) shall be accomplished by completely enclosing the item(s) in ferrous material or ferritic compositions of sufficient thickness to prevent item degradation.

5.2.2.6 Radioactivity protection. Protection against radioactivity shall be accomplished by completely enclosing the item in lead or lead filled compositions of sufficient thickness to prevent item degradation.

5.2.2.7 Marking. In addition to the standard markings required by MIL-STD-129 and the contract or order, the MIL-STD-129 markings for sensitive electronic items shall apply except as follows: The sensitive electronic device caution label, 2 inch by 2 inch or less, specified for intermediate packages shall be used for unit packages in lieu of the device symbol. In addition, special caution markings shall be placed on all unit and intermediate

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(interior) packages and shipping containers. Electrostatic sensitive item caution markings may be placed adjacent to the MIL-STD-129 sensitive electronic device caution label. Caution markings shall be as follows:

"CAUTION--ELECTROSTATIC SENSITIVE DEVICE: Remove electrostatic protection at use or in protected area. Reuse packaging materials for the unserviceable item. See NAVSEA 0D46363 for protective handling or testing measures for this item."

5.3 Cushioning, dunnage, and wrapping materials.

5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler or dunnage is prohibited for materials destined for shipboard installation/stowage.

5.3.2 Level C preservation-packaging and packing. When loose fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate and shipping) shall be marked or labelled with the following information:

"CAUTION

Contents cushioned etc., with loose-fill material. Not to be taken aboard ship. Remove and discard loose-fill material before shipboard stowage. If required, recushion with celluiosic material, bound fiber, fiberboard or transparent flexible cellular material."

5.3.3 Cushioning, filler, dunnage and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire.

6. NOTES

6.1 Intended use. The UPS covered by this specification is intended to provide uninterruptible Navy shipboard 60-Hz electrical power for vital loads.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Rating (see 3.3.3).
(c) Set of interim repair parts (see 3.8.2).
(d) Quality assurance requirements (see 4.1.3).
(e) Whether first article inspection is required (see 4.4).
(f) Level of preservation-packaging and packing required (see 5.1).
(g) Quantity of repair parts other than specified (see 5.2).
(h) Packaging other than MS90363 (see 5.2.1).

6.2.2 Data requirements. When this specification is used in a contract which invokes the provision of the "Requirements for Data" of the Defense Acquisition Regulation (DAR), the data identified below, which are required to be developed by the contractor, as specified on an approved Data Item Description (DD Form 1664), and which are required to be delivered to the Government, should be selected and specified on the approved Contract Data Requirement List (DD Form 1423) and incorporated in the contract. When the provisions of the "Requirements for Data" of the DAR are not invoked in a contract, the data required to be developed by the contractor and required to be delivered to the Government should be selected from the list below and specified in the contract.

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Paragraph	Data requirements	Applicable DID	Option
3.4.8	Safety statement	UDI-S-23097	------
3.4.8.1 and 3.4.8.2	Report, system hazards analyses	UDI-H-23391	------
3.7.4	Plan, reliability program	DI-R-2113	------
3.7.4.1.1	Failure data collection analysis and corrective action reporting program plan	UDI-R-20336	------
3.7.4.1.3	Report, reliability prediction	DI-R-2117	------
3.7.4.1.4	Report, failure mode effects and analysis (PMEA)	DI-R-2115	------
3.7.4.2 and 3.7.4.2.1	Plan, maintainability program	DI-R-2127	------
3.7.4.2.3	Reports, maintenance engineering analysis (MEA)	UDI-L-26377	------
3.7.5	Report, reliability and maintainability allocations, assessments, and analysis	UDI-R-23570	------
3.9.1, 3.9.1.1, 3.9.1.2, 3.9.1.3, and 3.9.1.4	Drawings, engineering and associated lists	DI-E-7031	Level 2 Design activity designation - contractor Design number - contractor
3.9.2	Manual, technical, functionally oriented maintenance manual (FOMM)	DI-M-2045	Type I of MIL-M-24100
4.4.1	Report, operational/performance test	UDI-T-23740	------
4.7.36 and 4.7.37	Report, component shop noise test	UDI-T-23764	------
4.7.39.1	Procedures, test	UDI-T-23732	------
5.1	Drawings, engineering and associated lists	DI-E-7031	Level 2 required for standard packages Level 3 required for reusable containers Design activity designation - contractor Design number - contractor
5.1	Preservation-packaging and packing test report	UDI-T-23766	------
5.2.1	Packaging and transportation support data	UDI-P-23508	------

(Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.2.2.1 The data requirements of 6.2.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.3 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or

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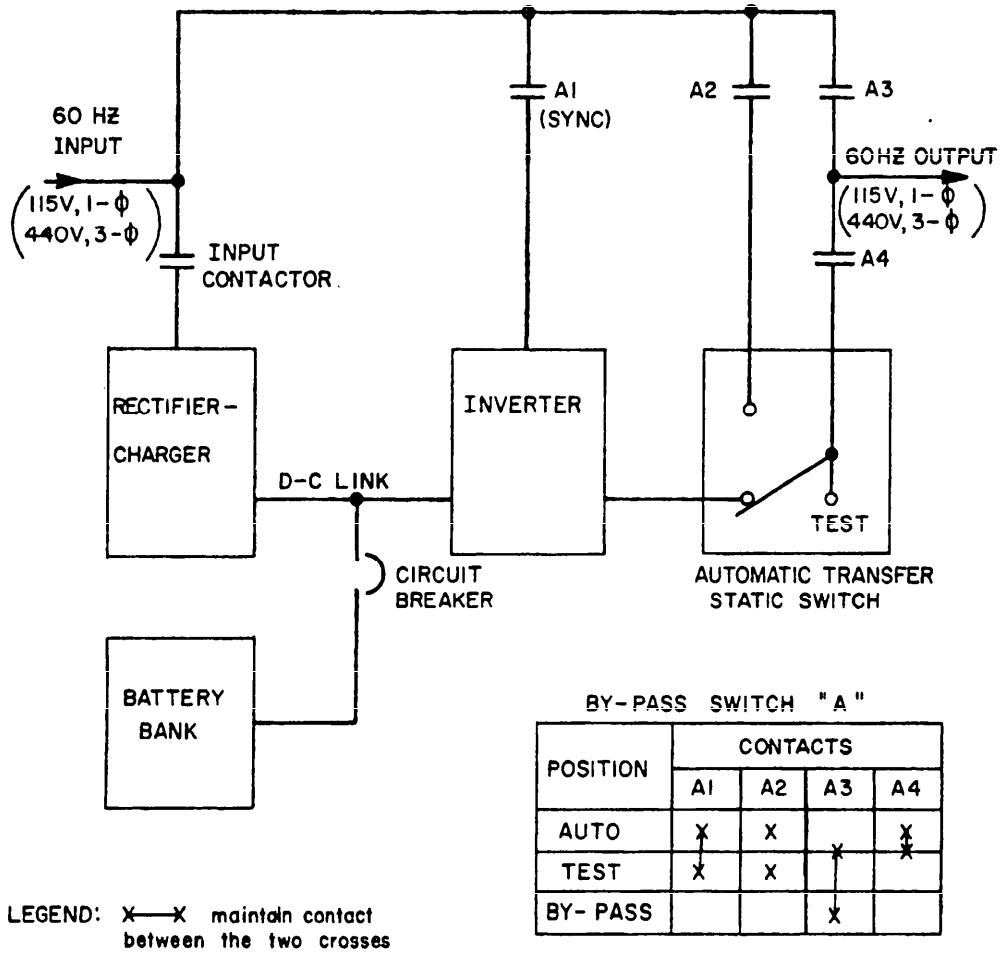
test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. When a manufacturer can present satisfactory test data previously taken on identical power rating unit and accompanied by a NAVSEC acceptance letter, the contracting agency may waive the first article inspection requirements provided that the test data submitted were taken within the past five years.

6.4 Definition. The UPS is defined in accordance with DOD-STD-1399, Section 300.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
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BY-PASS SWITCH "A"

POSITION	CONTACTS			
	A1	A2	A3	A4
AUTO	X	X		X
TEST	X	X	X	X
BY-PASS			X	

SH 11330

FIGURE 1. Basic UPS configuration.

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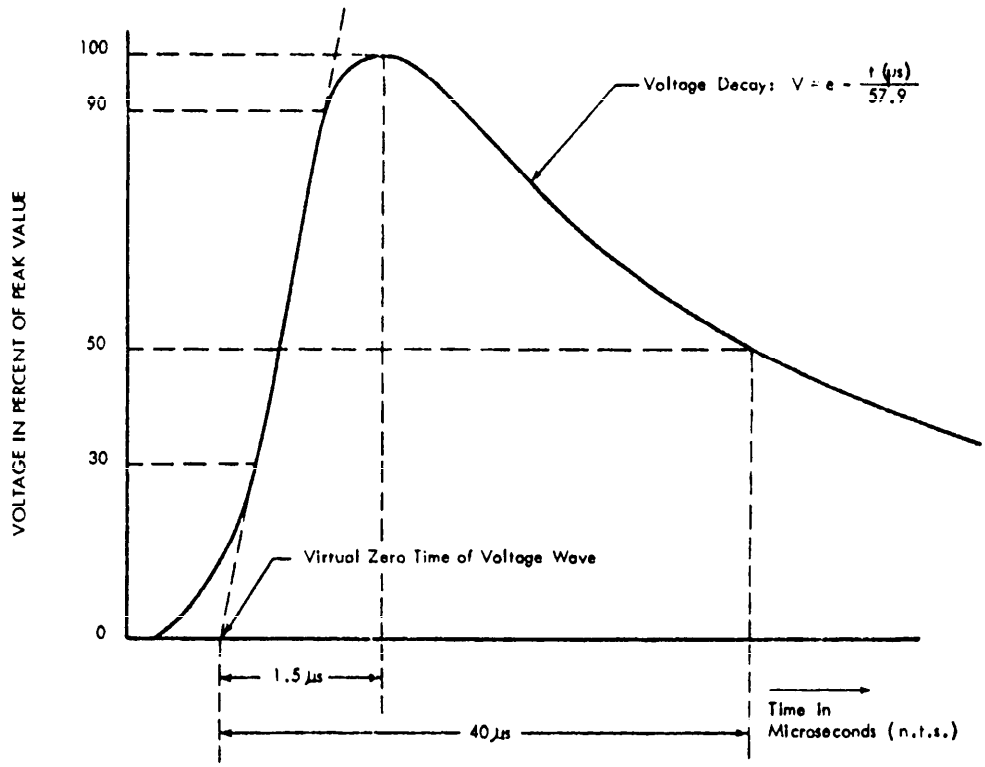
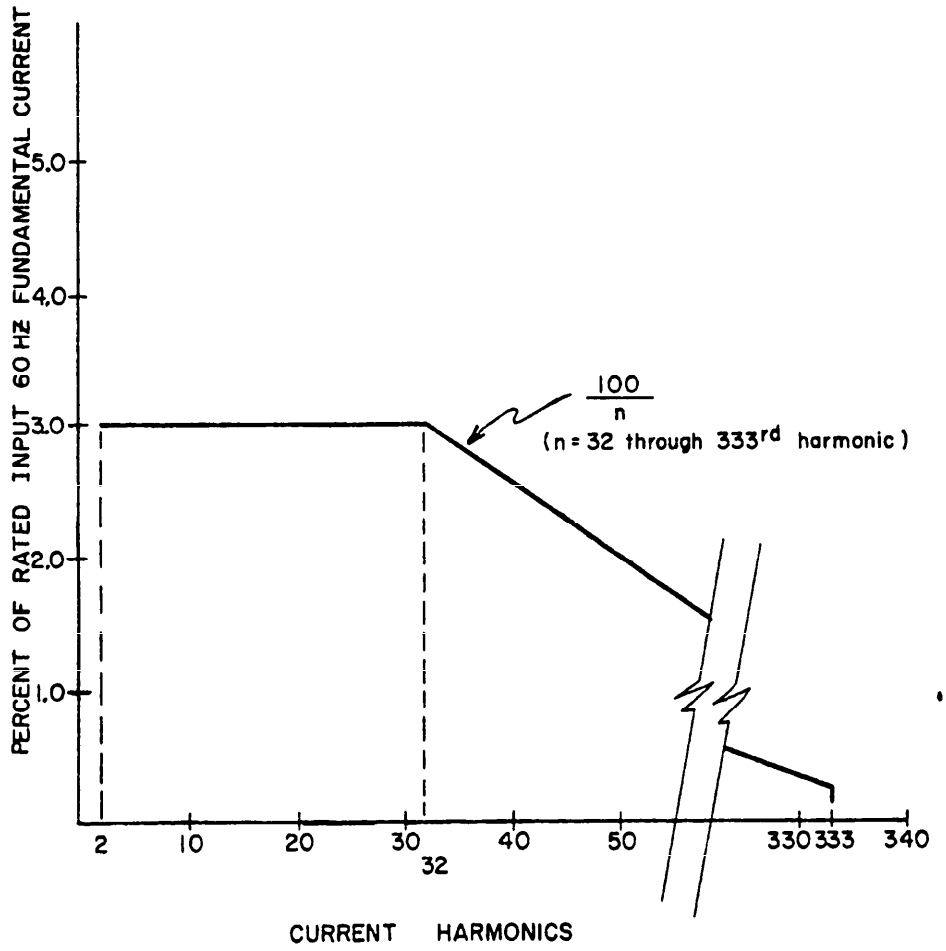


FIGURE 2. Spike voltage (short time transient) wave shape.

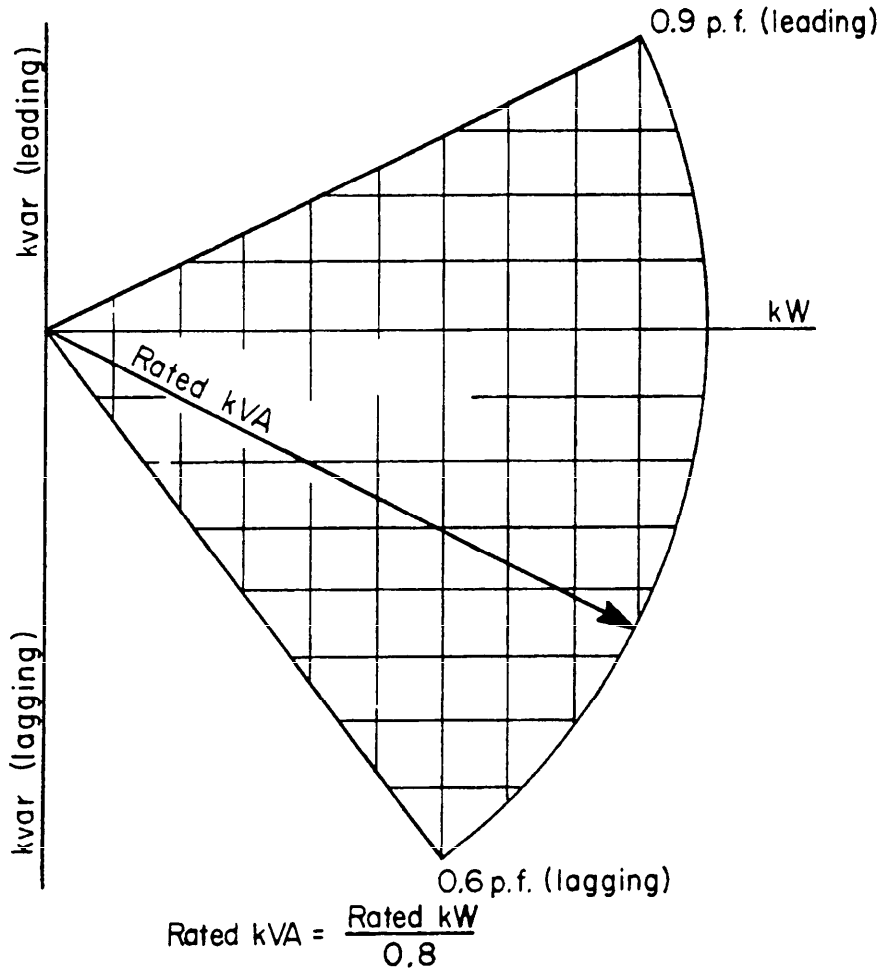
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SH 11331

FIGURE 3. Maximum allowable input current harmonics.

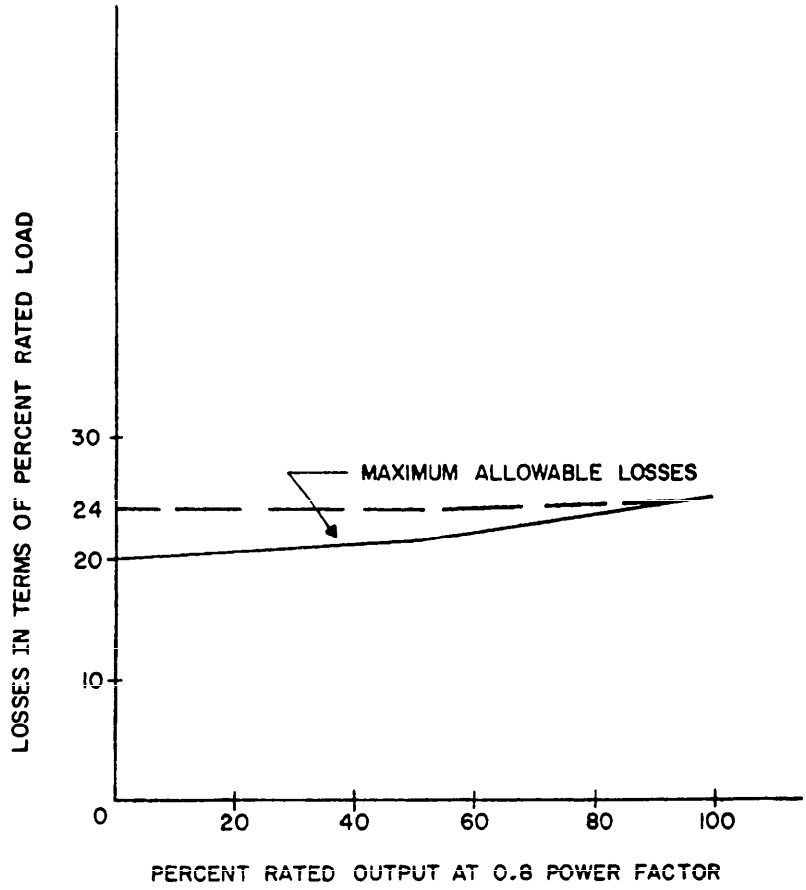
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SH 11285

FIGURE 4. Rating.

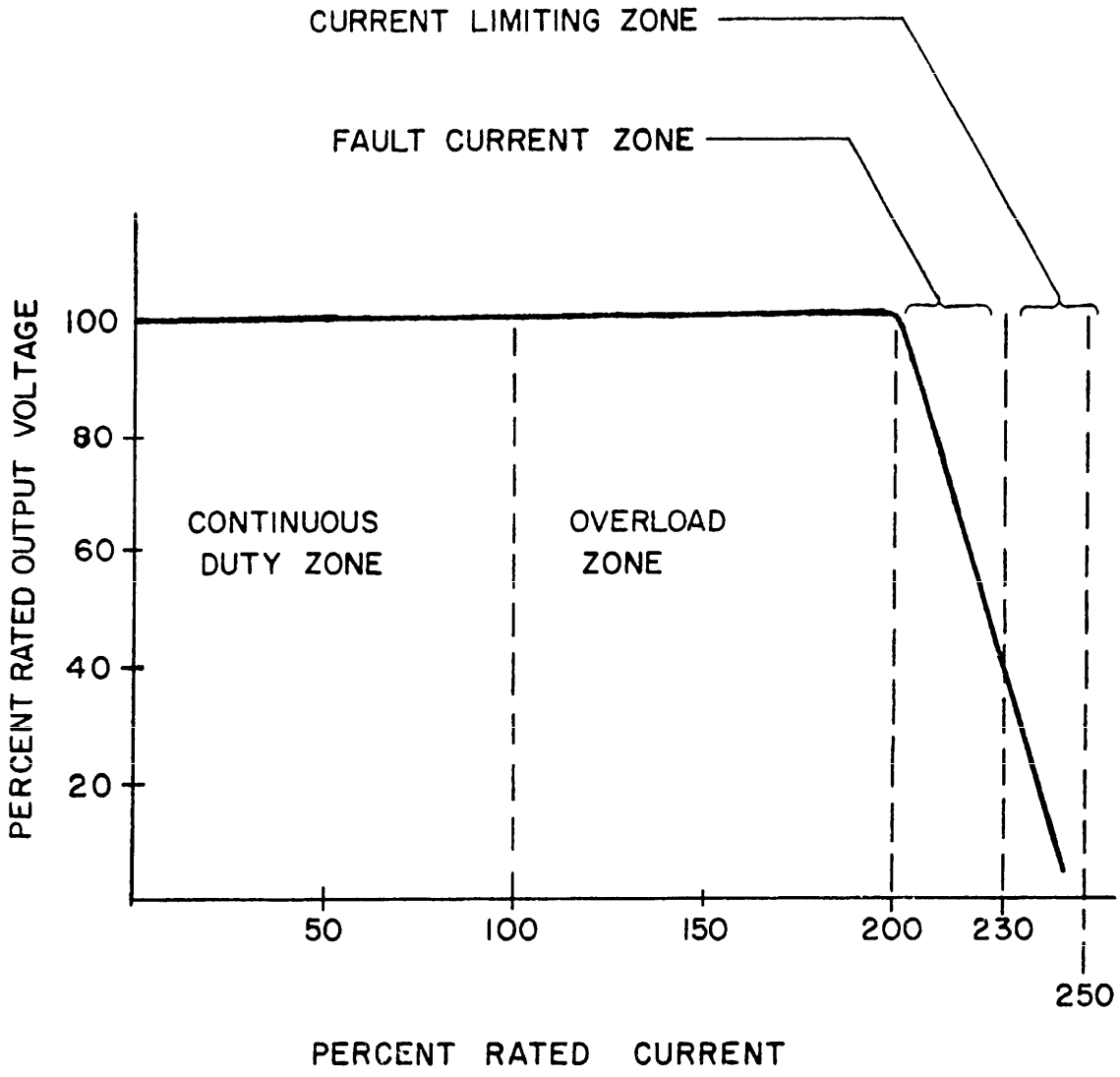
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SH 11332

FIGURE 5. Maximum allowable losses.

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SH 11333

FIGURE 6. Voltage versus current.

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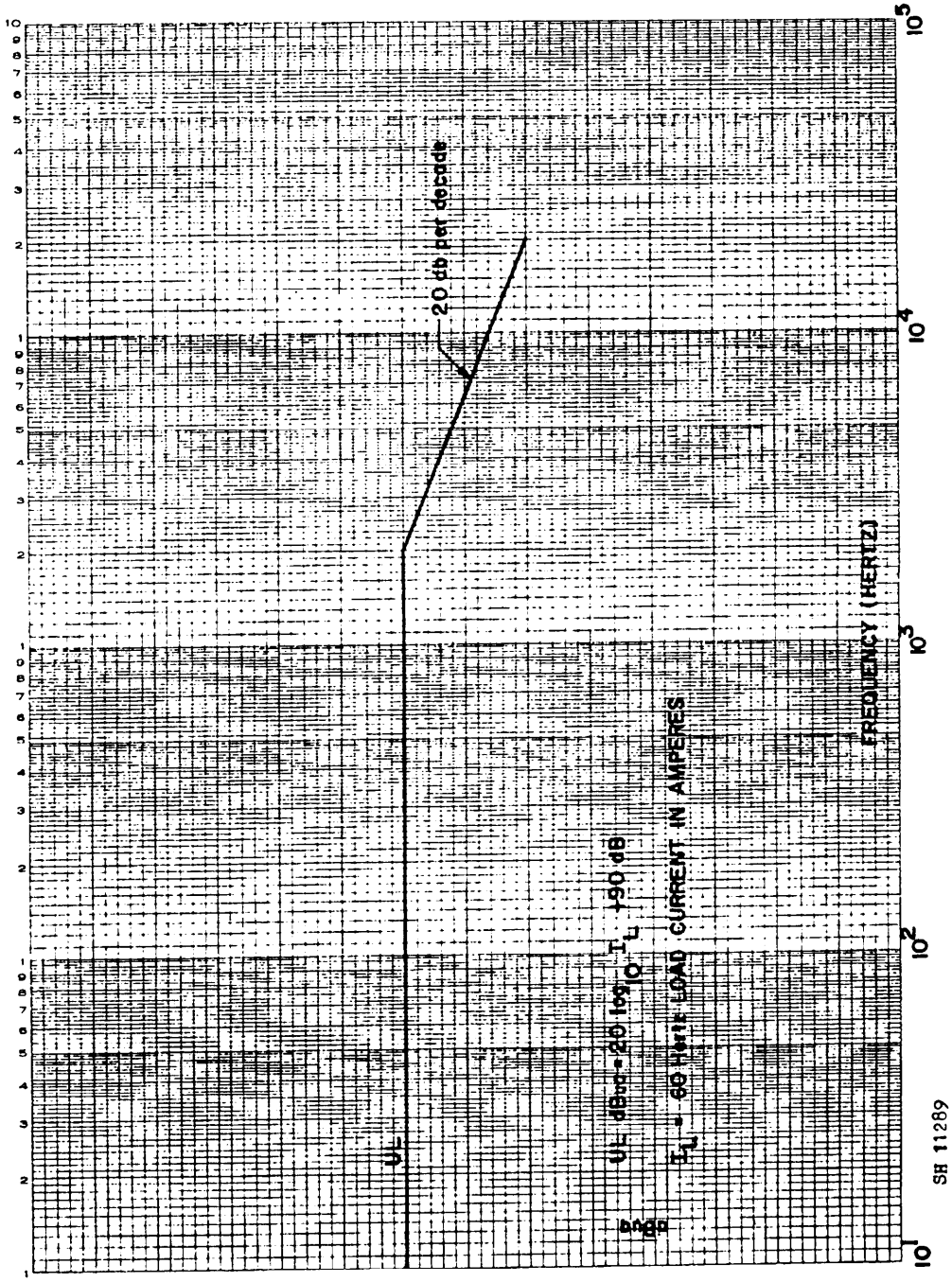


FIGURE 7. CE01 narrowband 60 Hz limit.

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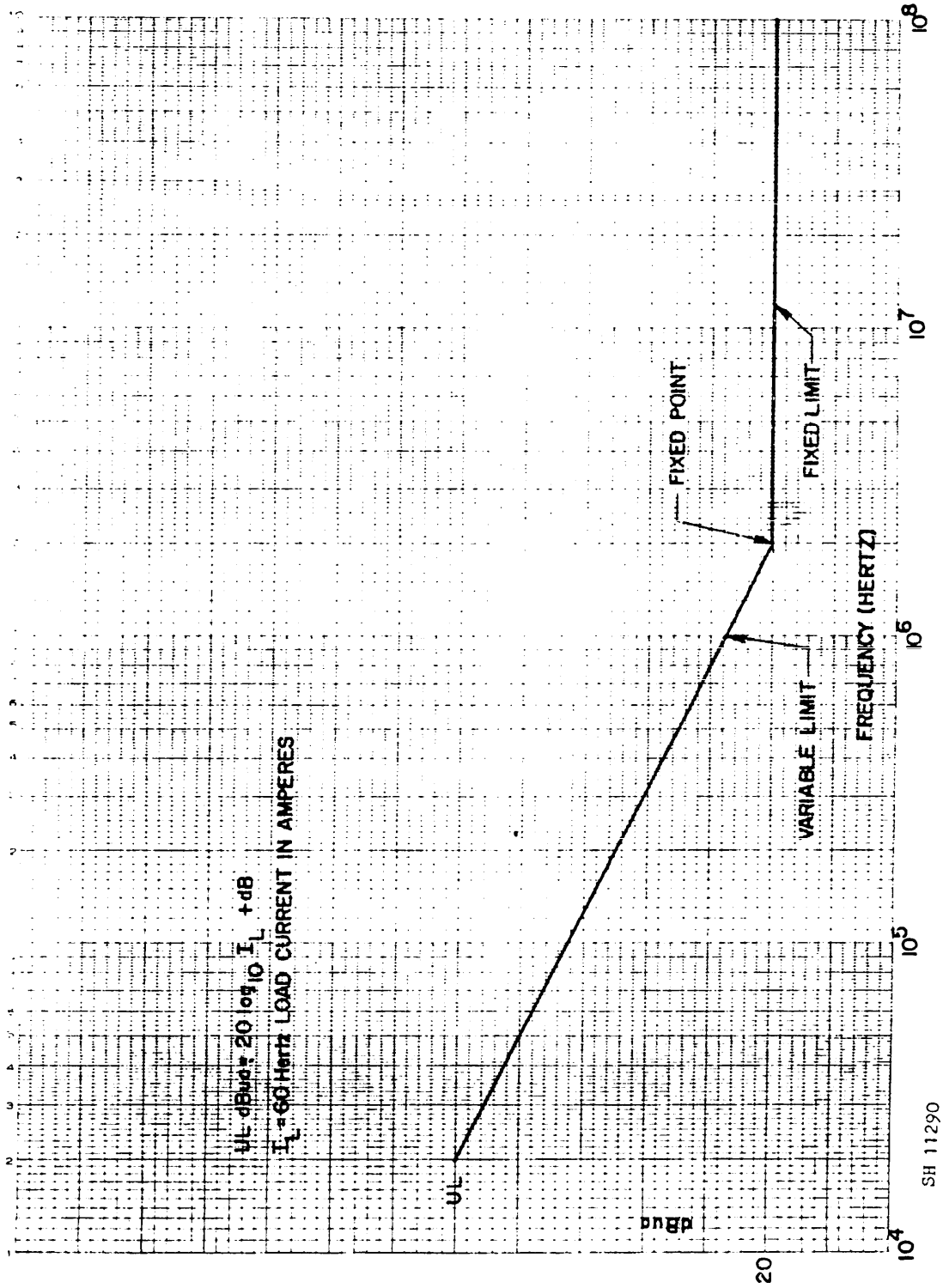
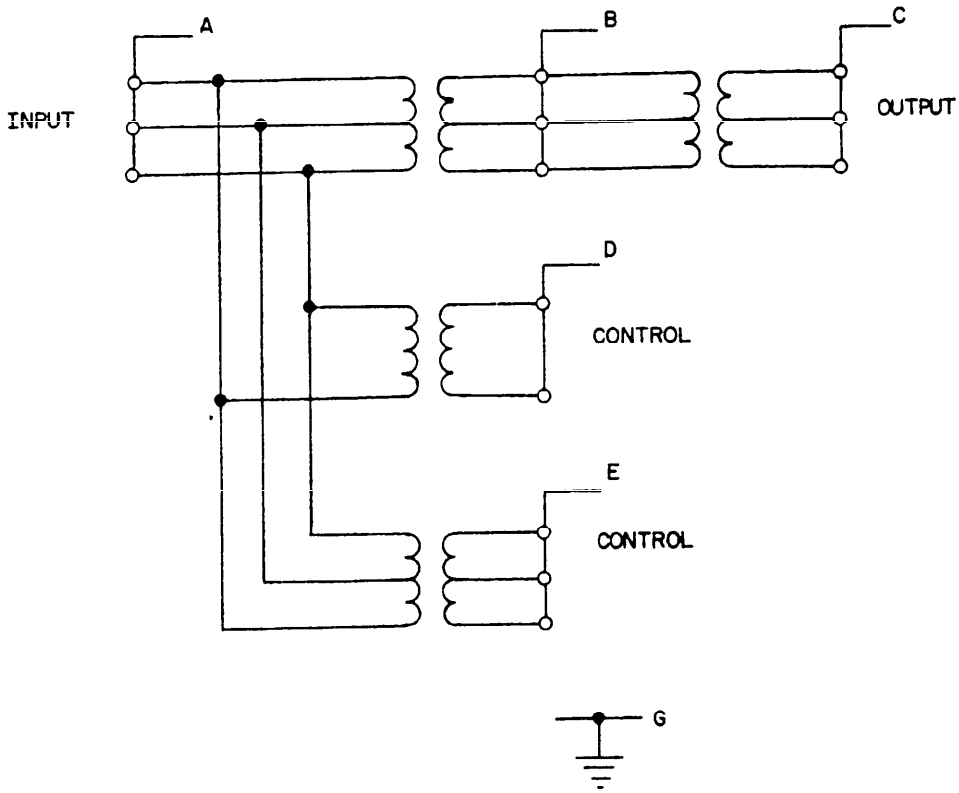


FIGURE 8. CE03 narrowband 60 Hz limit.

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MINIMUM NUMBER OF MEASUREMENTS REQUIRED:

1. A to G, B, C, D, and E connected.
2. B to G, A, C, D, and E connected.
3. C to G, A, B, D, and E connected.
4. D to G, A, B, C, and E connected.
5. E to G, A, B, C, and D connected.

SH 11334

FIGURE 9. Circuit connections for insulation resistance and dielectric withstanding voltage tests.

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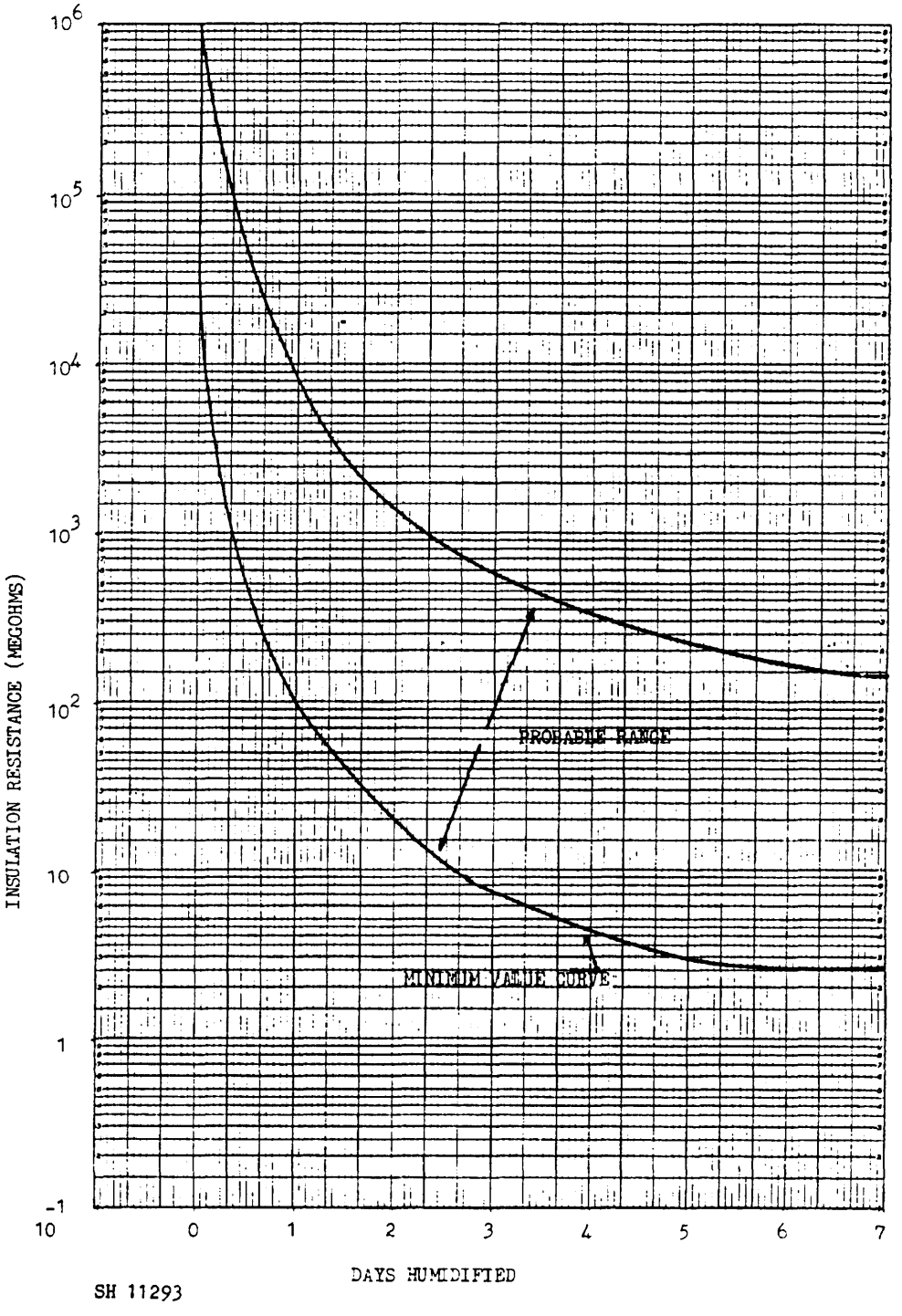


FIGURE 10. Insulation resistance.

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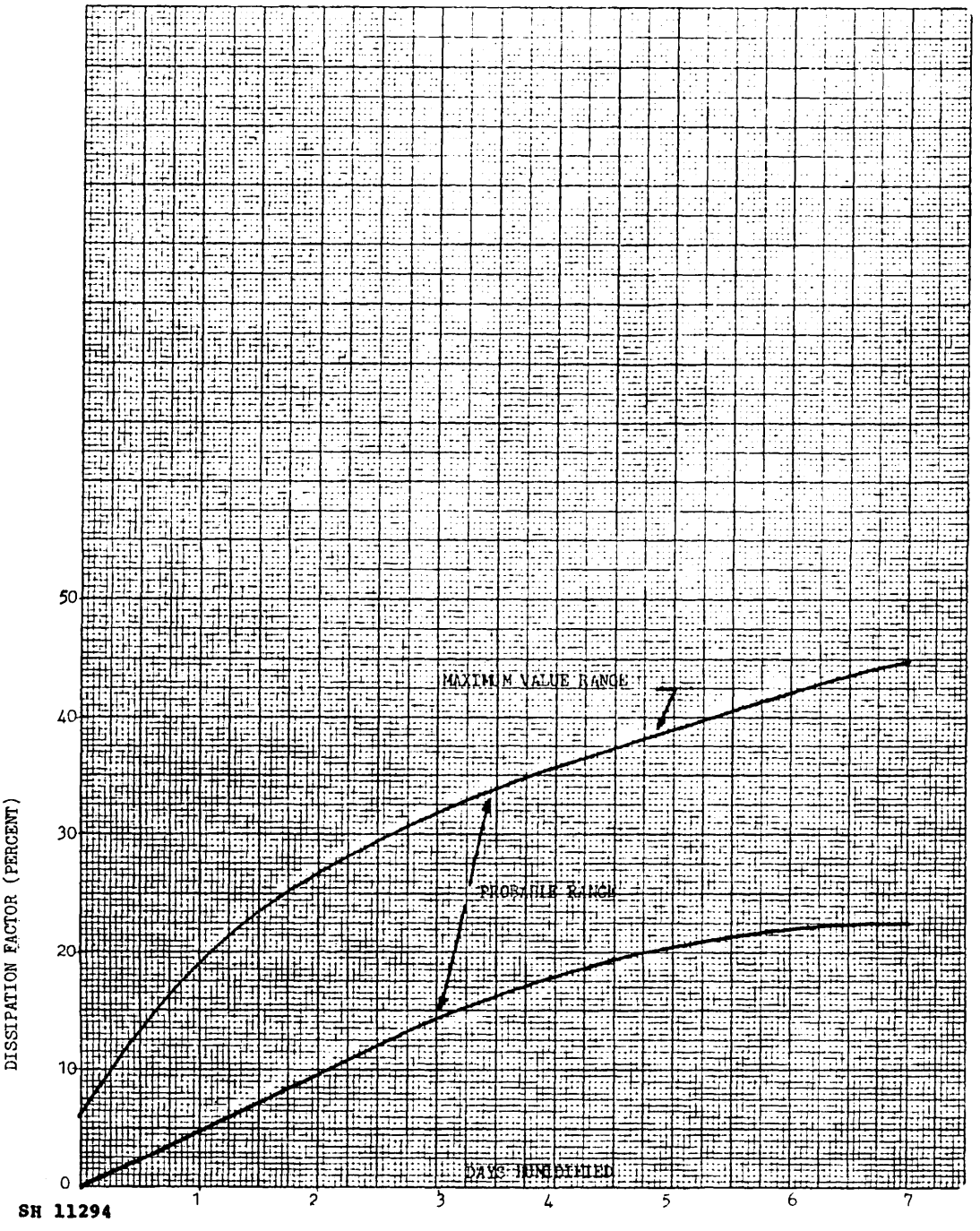


FIGURE 11. Dissipation factor.

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